



Transmission Expansion Advisory Committee (TEAC) Recommendations to the PJM Board

PJM Staff White Paper

PJM Interconnection
February 2025

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I. Executive Summary

On Aug. 7, 2024, the PJM Board of Managers approved changes to the Regional Transmission Expansion Plan (RTEP), totaling a net increase of \$618.06 million for baseline projects to resolve baseline reliability criteria violations and address changes to existing projects.

Since then, PJM has identified new baseline reliability criteria violations and the transmission system enhancements needed to solve them, at an estimated cost of \$5,919.77 million. Scope and cost changes to existing projects will result in a net increase of \$782.94 million, and cancellation to existing projects will result in a net decrease of \$48.7 million. This yields an overall RTEP net increase of approximately \$6,654.01 million to resolve baseline criteria violations, for which PJM recommended Board approval. PJM is also providing the 2024 annual update of RTEP generation and merchant transmission Network Upgrades in this white paper. New projects with signed Generation Interconnection Agreements (GIAs), project scope changes and project cancellations have resulted in a net increase of \$1,078.58 million for Network Upgrades. With these changes, RTEP projects will total approximately \$57,803.64 million since the first Board approvals in 2000.

PJM sought the Reliability and Security Committee consideration and full Board approval of the RTEP baseline projects summarized in this white paper. On February 26, 2025 the Board approved the addition of RTEP baseline projects as well as other changes to the RTEP as summarized in this paper.

II. Baseline Project Recommendations

A key dimension of PJM's RTEP process is baseline reliability evaluation, which is necessary before subsequent interconnection requests can be analyzed. Baseline analysis identifies system violations to reliability criteria and standards, determines the potential to improve the market efficiency and operational performance of the system, and incorporates any public policy requirements. PJM then develops transmission system enhancements to solve identified violations and reviews them with stakeholders through the Transmission Expansion Advisory Committee (TEAC) and subregional RTEP committees prior to submitting its recommendation to the Board. Baseline transmission enhancement costs are allocated to PJM responsible customers.

III. Baseline Reliability Projects Summary

A complete listing of all recommended projects and their associated cost allocations is included in Attachment A (allocations to a single zone) and Attachment B (allocations to multiple zones). The recommended projects include both noncompetitive solutions and competitive solutions stemming from the 2024 RTEP Window 1.

Detailed descriptions of the 2024 RTEP Window 1 solutions that PJM recommended to the Board are included in the [2024 RTEP Window 1 Reliability Analysis Report](#) (PDF) and the [2024 RTEP Window 1 Constructability & Financial Analysis Report](#) (PDF).

IV. Transmission Owner Criteria Projects

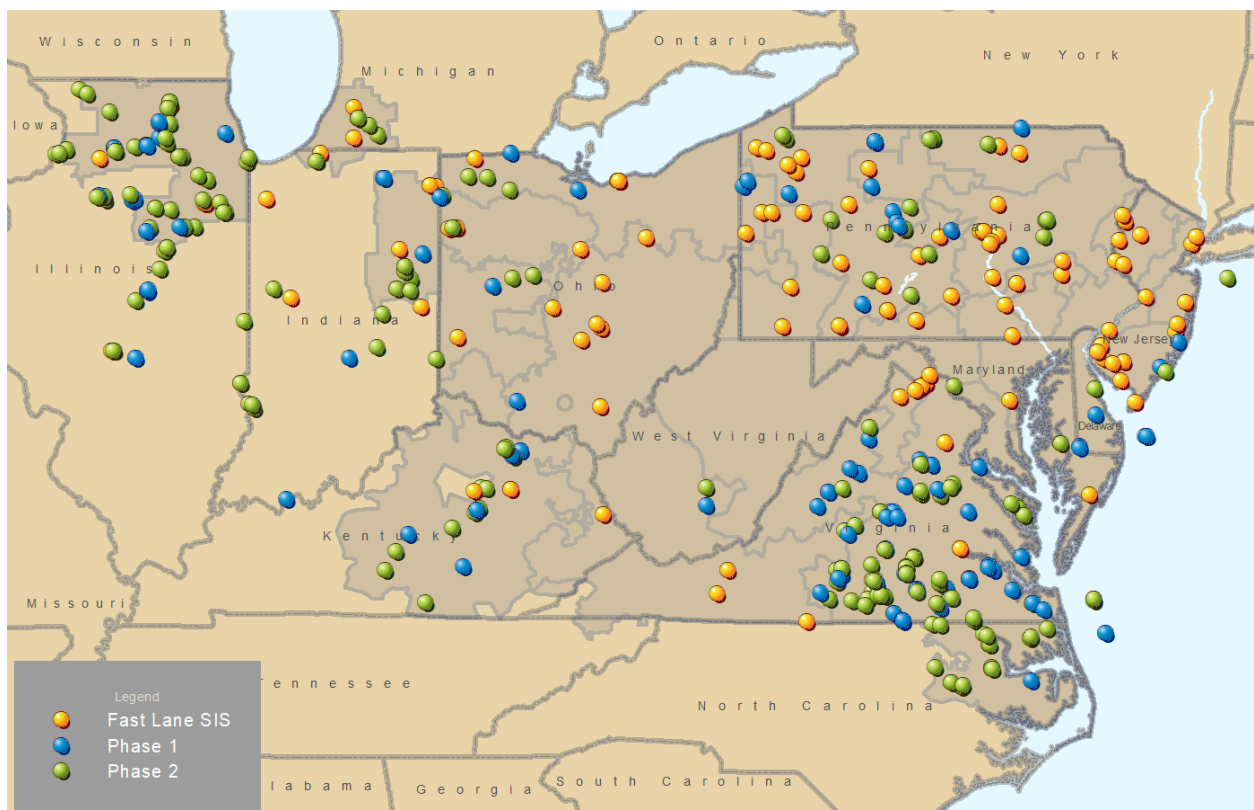
Of the \$5,919.77 million of new recommended baseline transmission system enhancements, approximately \$297.12 million is driven by transmission owner planning criteria, which makes up approximately 5% of the new project cost estimates.

V. New Services Queue

Throughout 2024, PJM has continued to study New Service Requests. These studies evaluate the impact of the New Service Requests and include an evaluation of new generation interconnections, increases in generation at existing stations, long-term firm transmission service requests and merchant transmission interconnection requests.

These studies were last reviewed with the Board Reliability Committee in December of 2023. Since that time, PJM has completed a total of 424 System Impact Studies (113 under the serial process and 311 for Phase I and Phase II of Transition Cycle #1), and 494 service requests have withdrawn. New projects with signed Interconnection Service Agreements (ISAs), project scope changes and project cancellations have resulted in a net increase of approximately \$1,078.58 million for Network Upgrades. The map below shows the locations of the new units associated with the completed interconnection System Impact Studies along with the fuel type and relative size. A listing of the projects with recently completed impact studies is provided in Attachment C to this white paper. A listing of the Network Upgrades associated with these projects is shown in Attachment D to this report. The cost for the Network Upgrades associated with these interconnection projects is the responsibility of the developer.

Map 1. Completed Interconnection System Impact Studies



VI. Changes to Previously Approved Projects

Scope/Cost Changes

The following scope/cost modifications were recommended:

New Jersey State Agreement Approach Project:

- The New Jersey Board of Public Utilities approved a scope reduction by an order issued in December 2024 (Docket No. QO20100630). The scope change removes the prebuild extension work from baseline project b3737.22 [Larrabee Collector station scope of the New Jersey State Agreement Approach (SAA) project]. Prebuild extension work included the civil work (i.e., duct banks) to accommodate four HVDC circuits from the prebuild “point of demarcation” to each offshore wind generator’s converter station area on the Larrabee Collector station property. The cost estimate for baseline project b3737.22 has decreased from \$217.1 million to \$208.9 million, resulting in a cost decrease of \$8.2 million.
- Additionally, the baseline project b3737.41 (Windsor to Clarksville subproject: terminal equipment upgrade at Clarksville 230 kV) is no longer required, as the Clarksville inside plant work was canceled. This reduction in scope yields a cost decrease of \$1.49 million.
- The total scope reduction for the New Jersey SAA project results in a net decrease of \$9.69 million.

Line No. 23 Bell Ave-Suffolk 115 kV Partial Rebuild:

The baseline project b3759 (line No. 23 segment Oak Ridge-Windsor PV1 115 kV reconductor) requires additional scope. The new scope of work wrecks and rebuilds the approximately 10.5 miles of 115 kV line No. 23 segment from Oak Ridge to Windsor PV1 tap (structures to be rebuilt to current 115 kV standards). Additionally, the 115 kV line No. 44 segment from Oak Ridge to Windsor PV1 tap, which also shares the same structures, will be reconducted. During the transmission line conceptual review of the original project scope, a number of issues were encountered for the majority of the transmission towers including:

- Inability to maintain adequate Dominion clearance using the new conductor
- ~80% of the towers will need grillage foundation modifications.
- ~50% of the structures experienced failures in two of the steel angle members located in the arms when subjected to the region's specified ice and wind parameters. Upgrading these structures by replacing the angles in the arms would necessitate full access to the structures. This includes the access roads, as well as equipment and labor costs to hold the wire in the air during arm modifications.

For these reasons, the modified scope of work is recommended. The cost estimate for this project has increased from \$23.5 million to \$41 million, resulting in a net increase of \$17.5 million.

Brandon Shores Deactivation Project:

The cost estimate for the Brandon Shore deactivation project, which includes baseline projects b3780 and b3781, has increased due to the following reasons:

- Actual vendor quotes received through the competitive bid process for major long-lead equipment were higher than budgetary quotes originally received during the conceptual design phase.
- Detailed 30% design- and engineering-identified necessary project changes after performing more detailed assessment of site conditions and engineering studies – i.e., New Batavia substation will change from AIS (air-insulated substation) to GIS (gas-insulated substation) due to space constraints and wetland conditions.
- Constructability reviews were conducted as part of 30% design and engineering to limit ad hoc changes during construction. The outcome of these efforts resulted in a more comprehensive plan for ROW (right-of-way) access and construction, which should limit cost overruns later in the project life cycle, and more importantly, last-minute changes to already challenging outage requirements.
- Additional contracting resources are required due to volume and scale of work on the system and outage constraints, which could require out-of-state resources to support without compromising existing projects and commitments. Labor rates for construction and engineering have also increased significantly over the past 1–2 years.
- Increased indirect costs and overhead to manage significant volume and complexity of work

The total cost of the Brandon Shores deactivation project, which includes baseline projects b3780 and b3781, has increased from \$738.83 million to \$1,513.96 million, resulting in a net increase of approximately \$775.13 million. The scope of the Brandon Shores deactivation project has not changed.

All of the changes noted above result in a net RTEP cost increase of \$782.94 million.

Cancellations

The following cancellations were recommended:

Baseline Project b3721:

The baseline project b3721 (Avery-Hayes 138 kV line rebuild/reconductor) in the ATSI zone is no longer required, as the b3925.3 project (2024 RTEP Window 1, proposal 605) will supersede b3721 with higher ratings. This cancellation yields a decrease of \$10.4 million.

Baseline Project b2003:

The baseline project b2003 (Whippany-Montville 230 kV line) in the JCPL zone is being canceled. FirstEnergy has requested this project be canceled due to routing and permitting issues. PJM will verify that the original violations remain valid and will consider alternative solutions if needed. This cancellation yields a decrease of \$37.5 million.

Baseline Project b3295:

The baseline project b3295 in the AEP zone (install a 34.5 kV 9.6 MVAR cap bank at Grundy station) is no longer required. A portion of the load in the area was retired and is no longer served from the transmission system, and there are no voltage violations in the area accordingly. This cancellation yields a decrease of \$0.8 million.

All of the changes noted above result in a net RTEP decrease of \$48.7 million.

VII. Review by the Transmission Expansion Advisory Committee (TEAC)

The project needs and recommended solutions discussed in this report were reviewed with stakeholders during 2024 and 2025, most recently at the February 4, 2025, TEAC meeting. In addition to receiving verbal feedback and comments during the relevant TEAC meetings, PJM also requested that written comments be submitted to PJM to communicate any concerns with project recommendations. PJM received significant stakeholder feedback and follow-up inquiries, particularly related to the 2024 RTEP Window 1 project proposal 262 and the Brandon Shores deactivation b3780 and b3781 cost increase. PJM considered all feedback received in deciding which solutions to recommend to the PJM Board for approval. Additionally, PJM is currently engaging with stakeholders and responding to all received inquiries. PJM will be publishing to the PJM website a summary of the requests received together with PJM responses.

VIII. Cost Allocation

Cost allocations for recommended projects are shown in Attachment A (for allocation to a single zone) and Attachment B (for allocation to multiple zones).

Cost allocations are calculated in accordance with Schedule 12 of the Open Access Transmission Tariff. Baseline reliability project allocations are calculated using a distribution factor methodology that allocates cost to the load zones that contribute to the loading on the new facility. The allocations will be filed at FERC no later than 30 days following approval by the Board.

IX. Board Approval

The PJM Reliability and Security Committee is requested to endorse the additions and changes to the RTEP proposed in this white paper and to recommend to the full Board for approval the new projects and changes to the existing RTEP projects as detailed in this white paper. On February 26, 2025 the Board approved the addition of RTEP baseline projects as well as other changes to the RTEP as summarized in this paper.

Attachment A – Reliability Project Single-Zone Allocations

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b3857.1	Conastone 230 kV breaker No. 5	\$1.92	BGE	BGE (100.00%)	6/1/2029
b3858.1	Rebuild the Windsor substation 115 kV yard to convert from a straight bus configuration into a six-breaker ring bus configuration. Install two (2) 21.62 MVAR, 115 kV capacitor banks. The substation fence will need to be expanded and approximately one acre of land will need to be purchased. No wetlands or environmental risks were identified at this time. The Tolna and Yorkana 115 kV line exits will need to be relocated. Below Grade: - Install foundation, trench, conduit and grounding for new equipment. - Install fencing, stoning, grading, and ground grid for expansion area. - Install conduit for fiber. Above Grade: - Install (4) 115 kV, 3000 A 40 kAIC circuit breakers. - Relocate (2) 115 kV circuit breakers (B4 and B98932). - Install (12) 115 kV, 1200A GOAB disconnect switches. - Install (4) 115 kV, 1200 A MOAB disconnect switches. - Install (6) 115 kV CVTs. - Install (6) 115 kV surge arresters. - Install (4) 115 kV H-frame dead-ends. - Install (1) 115 kV wave trap. - Install (2) 115 kV, 21.62 MVAR cap banks. - Install (1) 115 kV SSVT. - Install (1) lot of cables, grounding, rigid bus and fittings for new equipment. - Install (1) lot of fencing. Relaying and Control: - Install (6) breaker control panels with (1) SEL-451. - Install (2) standard line relay panels with (1) SEL-421 and (1) SEL-411L. - Install (1) carrier panel. - Install (2) standard cap bank panels with (2) SEL-487Vs. - Install (1) lot of control cables, SEL cables and fiber. Additional Equipment To Be Removed: - Existing foundations, conduit, steel structures. (1) 115 kV circuit breaker (6) 115 kV disconnect switches, 115 kV relay panels (1) 115 kV wave trap, line tuner and coax	\$12.36	ME	ME (100.00%)	6/1/2028

b3858.2	Rebuild the Yorkana substation 115 kV yard converting from a straight bus configuration to a (9)-breaker, breaker-and-a-half configuration. The substation fence will need to be expanded, but no land acquisition will be required. The Glades, Windsor and Redfront 115 kV line exits will need to be relocated. Below Grade: • Install (1 Lot) of foundations, conduit and grounding for new equipment. • Install (1 Lot) of cable trench in 115 kV area of the yard. • Install (1 Lot) of stoning, grading and ground grid for the substation yard expansion. Above Grade: • Install (1) prefabricated control building with battery system, AC and DC aux power panels, security cabinet, and MPLS network equipment. • Install (1 Lot) of fencing for substation yard expansion. • Install (3) 115 kV A-frame dead-ends. • Install (3) 115 kV H-frame dead-ends. • Install (2) 115 kV monopole structures. • Install (9) 115 kV circuit breakers. • Install (18) 115 kV GOAB disconnect switches. • Install (6) 115 kV MOAB disconnect switches. • Install (18) 115 kV surge arresters. • Install (18) 115 kV CVTs. • Install (2) 115 kV SSVTs. • Install (1 Lot) of rigid bus, insulators, fittings, conductor, connectors and support structures. Relaying and Control: • Install (3) standard FE line relay panels, consisting of (1) SEL-421, (1) SEL-411L, over fiber. • Install (9) standard FE breaker panels, consisting of (1) SEL-451. • Install (3) standard FE transformer panels. • Install (2) standard FE bus diff panels. • Install (1) standard FE cap bank panels. • Install (1) SCADA RTU. • Install (1) HMI panel with GPS clock and RTAC. • Install (1) ATS. • Install (1 Lot) of control cables and SEL cables. Additional Equipment To Be Removed: • Remove existing 115 kV rigid bus, steel. • (7) Circuit breakers • (12) Disconnect switches • (10) CVTs • (12) Surge arresters	\$20.80	ME	ME (100.00%)	6/1/2028
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Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
	(5) A-frame dead-ends Remove all existing 115 kV relaying and controls: (3) Transformer panels (2) Bus panels (1) Cap bank panel (3) Line panels (7) Breaker panels				
b3860.1	Replace the 34.5 kV bus tie breaker at Chester substation with a new 34.5 kV breaker that has an interruption capability of 40 kA.	\$0.54	JCPL	JCPL (100.00%)	6/1/2029
b3861.1	Replace the W101 34.5 kV breaker at Werner substation with a new 34.5 kV breaker that has an interruption capability of 40 kA.	\$0.54	JCPL	JCPL (100.00%)	6/1/2029
b3862.1	Upgrade two (2) 500 kV free standing CTs, one (1) disconnect switch, and four (4) sections of tube bus at Elroy 500 kV substation.	\$1.50	PECO	PECO (100.00%)	6/1/2029
b3863.1	Change tap ratios on two (2) CTs at Schuylkill 230 kV substation.	\$0.10	PECO	PECO (100.00%)	12/1/2029
b3864.1	Replace station cable at Tacony 69 kV station.	\$0.18	PECO	PECO (100.00%)	6/1/2029
b3864.2	Replace station cable at Richmond 230 kV station.	\$0.18	PECO	PECO (100.00%)	6/1/2029
b3865.1	Upgrade 138/69 kV autotransformer, a 69 kV breaker, two disconnects and move a takeoff structure at Reybold substation. These upgrades will require a substation expansion to move the takeoff structure and a control house expansion to move the 69 kV breaker relays from the Delaware City control house to the Reybold control house.	\$9.48	DPL	DPL (100.00%)	12/1/2029
b3866.1	Adjust relay setting at Cedar Creek 230 kV substation.	\$0.10	DPL	DPL (100.00%)	12/31/2029
b3866.2	Change relay setting at Milford 230 kV substation	\$0.10	DPL	DPL (100.00%)	12/31/2029
b3867.1	Upgrade a circuit switcher at Mickleton substation.	\$1.03	AEC	AEC (100.00%)	6/1/2029
b3868.1	Cut existing Carlstadt to River Road 69 kV line and extend Carlstadt line side to Penhorn 69 kV. Extend the other end of the line by constructing a new portion and connecting it to Kingsland 69 kV switch.	\$14.62	PSEG	PSEG (100.00%)	6/1/2029
b3868.2	Extend the other end of L-636 to Kingsland switch by constructing new 5.5-mile portion utilizing existing I-2314 transmission towers from H-A 5/4 to H-A 2/3. New 69 kV line to be routed along County Ave. pass Secaucus Rd. in Secaucus NJ.	\$30.87	PSEG	PSEG (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b3868.3	Reconfigure former River Road to Carlstadt 69 kV and Tonnelle Ave. to Union City 69 kV lines at the intersection Tonnelle Ave. and Granton Ave. in North Bergen, NJ by connecting Union City to River Road and Tonnelle Ave. to Kingsland.	\$0.51	PSEG	PSEG (100.00%)	6/1/2029
b3869.1	Relocate the Bergen Gen No. 1 point of interconnection from Bergen 138 kV to Bergen 345 kV GIS through the existing 345/138 kV transformer.	\$6.80	PSEG	PSEG (100.00%)	12/1/2029
b3869.2	Remove and retire the two (2) existing Bergen 138 kV series reactors and associated ancillary equipment.	\$5.70	PSEG	PSEG (100.00%)	12/1/2029
b3870.1	At Federal Street substation: • Install (3) 138 kV CCVTs on the 138 kV bus. • Remove the transformer 1 CO-6 138 kV phase relay overcurrent relays. • Install (1) SEL-421 relay on the high side of transformer 1. • Install foundations, conduit and grounding for new CCVTs. • Install cables between CCVTs and relay.	\$0.69	POTED	APS (100.00%)	6/1/2028
b3871.1	Replace two 138 kV disconnect switches with 3000 amp disconnect switches and replace a portion of the stranded conductor on the No. 2 138 kV bus with aluminum pipe bus.	\$8.00	DL	DL (100.00%)	6/1/2029
b3872.1	Adjust the tertiary tap on the Hartford 138/69/34.5 kV transformer 1 and on Hartford 138/69/12 kV transformer 4 to eliminate the high voltage issues and avoid circulating current.	\$0.10	AEP	AEP (100.00%)	4/15/2029
b3873.1	Install 12 MVAR 34.5 kV cap bank at Greenleaf.	\$1.31	AEP	AEP (100.00%)	6/1/2029
b3875.1	Reconductor ~3.95 miles of ACSR 6/1 Penguin (4/0) on the Firebrick-Jefferson switch 69 kV branch with ACSR 556.6 26/7. Remote end (line setting) would need to be updated at Firebrick and Lick. Replace 600A switches at Jefferson and replace 477 AA 19 sub conductor at Firebrick.	\$8.50	AEP	AEP (100.00%)	6/1/2029
b3876.1	Install a 69 kV 11.5 MVAR capacitor bank at Richlands station with a circuit switcher.	\$1.00	AEP	AEP (100.00%)	12/1/2029
b3877.1	Project will replace station conductor and switches in the 345 kV yard at Beatty that are currently limiting the 345 kV lines to Adkins and Chenoweth.	\$0.50	AEP	AEP (100.00%)	6/1/2029
b3877.2	Project will upgrade 345 kV circuit breakers 'A' and 'A1' to 4000A 63 kA breakers at Adkins station along with some station conductor that is currently limiting the 345 kV line to Beatty.	\$4.50	AEP	AEP (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b3878.1	Proposal will upgrade 765 kV circuit breakers 'B' and 'B2' to 5000A 50 kA breakers at Marysville station. In addition, the project will upgrade the existing wave trap toward Sorenson.	\$10.00	AEP	AEP (100.00%)	6/1/2029
b3879.1	Replace line conductor, ~0.11 miles of 4/0 ACSR 6/1 conductor with 556.5 26/7 between South Toronto and the South Toronto tap.	\$0.50	AEP	AEP (100.00%)	6/1/2029
b3879.2	Upgrade the wave trap, CCVTs, switches and station conductor at South Toronto station currently limiting the line to South Toronto tap.	\$0.60	AEP	AEP (100.00%)	6/1/2029
b3880.1	At Beatty Road substation, install a 69 kV, 23 MVAR capacitor bank along with the 69 kV cap bank breaker.	\$2.25	AEP	AEP (100.00%)	6/1/2029
b3882.1	Replace 138 kV circuit breaker BB with higher fault current capable counterpart.	\$1.00	AEP	AEP (100.00%)	6/1/2026
b3883.1	69 kV station equipment, including relays, conductor and switches, will be replaced at Haviland station in order to address identified overloads on the lines to North Van Wert and Cavett.	\$0.80	AEP	AEP (100.00%)	12/1/2029
b3884.1	Replace the 69 kV circuit breaker D at Van Wert with a 40 kA breaker.	\$1.00	AEP	AEP (100.00%)	6/1/2026
b3885.1	Replace 69 kV circuit breakers N and M at Schroyer Avenue station with higher fault current capable counterparts.	\$2.00	AEP	AEP (100.00%)	6/1/2026
b3886.1	Replace 69 kV circuit breaker 'A' along with disconnect switches at Benwood substation with a 40 kA circuit breaker.	\$1.00	AEP	AEP (100.00%)	6/1/2026
b3887.1	Replace Greentown 138 kV circuit switcher for transformer No. 5 with a 138 kV 63 kA circuit breaker.	\$1.00	AEP	AEP (100.00%)	6/1/2026
b3888.1	Preform sag study and complete mitigations on the 138 kV line between East Leipsic and the AE2-072 tap (Lammer) to allow line's conductor to operate to its maximum operating temperature (MOT).	\$2.50	AEP	AEP (100.00%)	6/1/2029
b3889.1	Project will replace limiting station equipment at Tiltonville station to increase the rating on the branch to Windsor.	\$0.80	AEP	AEP (100.00%)	6/1/2029
b3890.1	Replace station conductor at South Coshocton station currently limiting the branch to Ohio Central.	\$0.20	AEP	AEP (100.00%)	6/1/2029
b3891.1	Project will perform relay upgrades at Kenny 138 kV to raise the CT and relay thermal limits that are currently limiting the line to Roberts.	\$0.40	AEP	AEP (100.00%)	6/1/2029
b3892.1	Replace 69 kV circuit breakers A and S at Mount Vernon station with 40 kA breakers.	\$2.00	AEP	AEP (100.00%)	6/1/2026
b3894.1	Replace limiting station conductor at Tidd on the line to Carnegie (FE).	\$0.50	AEP	AEP (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b3895.1	Replace existing 138 kV, 40 kA circuit switcher L at Jacksons Ferry station with new 138 kV, 63 kA circuit breaker	\$1.00	AEP	AEP (100.00%)	6/1/2026
b3896.1	Adjust the capacitor bank voltage settings to allow the cap bank to operate as needed under N-1-1 scenarios.	\$0.10	AEP	AEP (100.00%)	12/1/2029
b3897.1	Replace the 138 kV, 40 kA circuit switcher XT8 with a 63 kA circuit breaker.	\$1.00	AEP	AEP (100.00%)	6/1/2026
b3898.1	Upgrade the CT thermal limit at Buchanan station on the Buchanan-Keen Mountain 138 kV line.	\$0.10	AEP	AEP (100.00%)	12/1/2029
b3899.1	Replace OVEC-owned station equipment at Kyger Creek to raise the rating of the Kyger Creek-Sporn 345 kV line. Equipment to be replaced includes station conductor and a wave trap at Kyger Creek.	\$0.80	OVEC	OVEC (100.00%)	6/1/2029
b3900.1	Replace 300 copper conductor with 1113 kcmil ACSR conductor on Kewanee 138 kV bus No. 1.	\$0.20	ComEd	ComEd (100.00%)	4/1/2029
b3901.1	Reconductor 138 kV line from Electric junction to W541.	\$1.38	ComEd	ComEd (100.00%)	6/1/2029
b3901.2	Replace 2" tubular bus, 2-500 KCMIL Cu and 1113 KCMIL ACSR/AA on bus 2 and line terminal of 138 kV L11106 at TSS 111 Electric junction with bus that meets or exceeds a minimum thermal capability of 2036/2236/2690A (487/534/643MVA) SN/SE/SLD.	\$0.28	ComEd	ComEd (100.00%)	6/1/2029
b3902.1	Replace the Wilton Center 345 kV BT 4-5 circuit breaker.	\$2.70	ComEd	ComEd (100.00%)	12/1/2029
b3903.1	Replace the Kewanee 138 kV L7411 circuit breaker with a SF6 63 kA circuit breaker.	\$1.10	ComEd	ComEd (100.00%)	6/1/2026
b3904.1	Rebuild and reconductor 7.7 miles of 69 kV line with our standard 1351 AAC conductor from Rockford substation to the POI.	\$25.00	Dayton	Dayton (100.00%)	4/15/2029
b3906.1	Construct new Marley Neck 115 kV substation. Marley Neck 115 kV portion will accommodate 10 breaker-and-a-half bays, with only 6 bays planned for initial service while accommodating 4 future bays. Two standard 230/115 kV transformers will be connected between the 230 and 115 kV equipment with appropriate isolation methods.	\$107.62	BGE	BGE (100.00%)	6/1/2029
b3908.1	Install one 80 MVAR 230 kV capacitor bank at Pocono 230 kV substation.	\$4.93	PPL	PPL (100.00%)	6/1/2029
b3910.1	Susquehanna T10 230 kV station reconfiguration. Break the existing Susquehanna-Glen Brook 230 kV line and loop it 0.2 miles in and out of the Susquehanna T10 230 kV station.	\$9.50	PPL	PPL (100.00%)	6/1/2029
b3911.1	Rebuild the existing 1.1-mile Canal-Gay 138 kV oil-filled, pipe-type underground line to address overloads on the existing cable utilizing 5000 MCM XLPE cable.	\$15.60	AEP	AEP (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b3912.1	Rebuild the existing 2.2-mile Canal-Mound St. 138 kV oil-filled, pipe-type underground line to address overloads on the existing cable utilizing 5000 MCM XLPE cable.	\$31.09	AEP	AEP (100.00%)	6/1/2029
b3913.1	Rebuild 138 kV line section between Beatty and White Road stations (approximately 4.5 miles). Update remote end relay settings as needed.	\$18.80	AEP	AEP (100.00%)	6/1/2029
b3913.2	Rebuild 138 kV line section between White Road and Cyprus stations (approximately 3.34 miles). Update remote end relay settings as needed.	\$14.31	AEP	AEP (100.00%)	6/1/2029
b3914.1	Change no-load tap of 2 autotransformers at Fisk from 338.25 kV to 346.5 kV.	\$0.03	ComEd	ComEd (100.00%)	4/15/2029
b3914.2	Change no-load tap of 4 autotransformers at Crawford from 338.25 kV to 345.6.5 kV.	\$0.07	ComEd	ComEd (100.00%)	4/15/2029
b3914.3	Change no-load tap of 3 autotransformers at Elmhurst from 338.25 kV to 346.5 kV.	\$0.05	ComEd	ComEd (100.00%)	4/15/2029
b3914.4	Change no-load tap of 2 autotransformers at West Loop from 338.25 kV to 346.5 kV.	\$0.03	ComEd	ComEd (100.00%)	4/15/2029
b3914.5	Change no-load tap of 2 autotransformers at Taylor from 338.25 kV to 346.5 kV.	\$0.03	ComEd	ComEd (100.00%)	4/15/2029
b3914.6	Change no-load tap of 4 autotransformers at Skokie from 338.25 kV to 345.6.5 kV.	\$0.07	ComEd	ComEd (100.00%)	4/15/2029
b3916.1	Install a new 420 MVA 345/138 kV autotransformer and associated 345 kV and 138 kV circuit breakers at Itasca substation.	\$14.31	ComEd	ComEd (100.00%)	6/1/2029
b3918.1	Add one additional breaker, a 2nd 138/69 kV transformer and replace five 69 kV breakers and four 138 kV breakers at Miami substation.	\$12.00	Dayton	Dayton (100.00%)	6/1/2029
b3919.2	Establish a 0.18-mile double circuit 345 kV line extension to cut the existing Hyatt-West Millersport 345 kV line in and out of Corridor station.	\$3.10	AEP	AEP (100.00%)	6/1/2029
b3919.3	Install three new 345 kV breakers at Corridor station in order to accommodate the cut in of the Hyatt-West Millersport 345 kV line.	\$5.52	AEP	AEP (100.00%)	6/1/2029
b3919.4	Reconductor 10.2 miles of Maliszewski-Corridor 345 kV line.	\$18.30	AEP	AEP (100.00%)	6/1/2029
b3919.5	Reconductor 4.75 miles of the existing Bokes Creek-Marysville 345 kV circuit. Update the associated relay settings.	\$6.67	AEP	AEP (100.00%)	6/1/2029
b3919.6	Rebuild 4.4 miles of the existing Marysville-Hyatt 345 kV double circuit line where it extends into Marysville station.	\$26.26	AEP	AEP (100.00%)	6/1/2029
b3919.7	Upgrade 345 kV breakers K and K1 along with associated switches and conductor to 5000A at Hyatt station.	\$2.94	AEP	AEP (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b3919.8	Upgrade the relaying and associated equipment at West Millersport station to coordinate with the cut-in work to Corridor station.	\$0.41	AEP	AEP (100.00%)	6/1/2029
b3919.9	Upgrade 3000A 345 kV breaker 'L2' along with associated terminal elements to 5000A at Marysville.	\$2.56	AEP	AEP (100.00%)	6/1/2029
b3919.10	Rebuild ~19.0 miles of Hyatt-Marysville 345 kV line using 4-bundled 795 ACSR conductor Bold construction (This is an EOL rebuild).	\$116.70	AEP	AEP (100.00%)	6/1/2029
b3921.1	Wreck and rebuild 115 kV line 119 from structure 119/305 (Merck No. 5 substation) to 119/411A (Port Republic substation). The existing structures shall be replaced one for one within the existing ROW using primarily custom engineered double circuit 115 kV steel structures on concrete foundations. The line will be rebuilt with 3-phase 1-768.2 ACSS/TW/HS (20/7) 250 MOT "Maumee" conductor and two (2) DNO-11410 OPGW. The rebuild includes the installation of double circuit structures but assumes the second circuit will not be installed as part of this project, and that the vacant conductor arms should not be utilized without acquiring additional ROW. This scope assumes project GITAE2029C will be completed prior to the construction of this project. Project GITAE2029C serves to install Port Republic substation, which will split line 119 in between existing structures 119/411 and 119/412.	\$44.84	VEPCO	Dominion (100.00%)	6/1/2029
b3921.2	Upgrade the 397.5 ACSR jumpers and associated equipment to meet the line conductor rating of 393MVA.	\$0.49	VEPCO	Dominion (100.00%)	6/1/2029
b3922.1	This project serves to wreck and rebuild 115 kV line 1031 from structure 1031/220 to structure 1031/329. The existing structures to be removed are primarily single circuit wood, steel or concrete monopoles. The existing structures to be removed were primarily constructed in 1993 with the weathering steel structures being constructed in 2011. See Figure 1 for the project location. The existing structures shall be replaced one for one within the existing ROW using single circuit steel monopoles on foundations. The line will be rebuilt with single circuit 3-phase 768.2 ACSS/TW/HS (20/7) "Maumee" conductor and single (1) DNO-11410 OPGW, respectively.	\$29.37	VEPCO	Dominion (100.00%)	6/1/2029
b3925.1	Rebuild the 7.46 miles of Avery-Shinrock 138 kV line with 795 kcmil 26/7 ACSS (7.46 miles).	\$15.20	ATSI	ATSI (100.00%)	6/1/2029
b3925.2	Rebuild the 13.45 miles of Greenfield-Lakeview 138 kV line from 2 x 336.4 kcmil 26/7 ACSR to 1 x 795 kcmil 26/7 ACSS.	\$59.68	ATSI	ATSI (100.00%)	6/1/2029
b3925.3	Rebuild 6.5 miles of Avery-Hayes 138 kV line with 795 kcmil 26/7 ACSS conductor.	\$11.01	ATSI	ATSI (100.00%)	6/1/2027

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b3925.4	Rebuild the Greenfield-Beaver 138 kV corridor (32 miles) with 795 kmil 26/7 ACSS. This corridor encompasses multiple 138 kV lines that are constructed on common towers.	\$131.43	ATSI	ATSI (100.00%)	6/1/2029
b3928.1	Install (1) 230 kV, 50 MVAR shunt capacitor bank and associated equipment including breaker at Navy North substation.	\$3.00	VEPCO	Dominion (100.00%)	6/1/2029
b3929.4	Rebuild approximately 7.7 miles of 230 kV line No. 2110 Suffolk-Thrasher that share the double circuit towers under Line No. 579.	\$21.25	VEPCO	Dominion (100.00%)	6/1/2029
b4000.100	Ashburn sub 230 kV – replace 50 kA breaker SC332 with 63 kA.	\$0.50	VEPCO	Dominion (100.00%)	6/1/2029
b4000.101	Beaumeade sub 230 kV – replace 63 kA breaker 274T2206 with 80 kA.	\$0.66	VEPCO	Dominion (100.00%)	6/1/2029
b4000.102	Braddock sub 230 kV – replace 40 kA breakers 207T294, 237T294, 237T297, 281T297 with 63 kA.	\$2.00	VEPCO	Dominion (100.00%)	6/1/2029
b4000.103	Brambleton sub 230 kV – replace 63 kA breakers 217202, 2172T2183, L102, L202 with 80 kA.	\$2.64	VEPCO	Dominion (100.00%)	6/1/2029
b4000.104	Bristers sub 230 kV – replace 40 kA and 50 kA breakers H1TH2, H2TH3, L1T2101 with 63 kA.	\$1.50	VEPCO	Dominion (100.00%)	6/1/2029
b4000.105	Bull Run sub 230 kV – replace 50 kA breaker H362 with 63 kA.	\$0.50	VEPCO	Dominion (100.00%)	6/1/2029
b4000.106	Buttermilk sub 230 kV – replace 63 kA breakers 215212, 217012, 220312, 221412, 2152T2203 with 80 kA.	\$3.30	VEPCO	Dominion (100.00%)	6/1/2029
b4000.107	Cabin Run sub 230 kV – replace 63 kA breakers 209512, 221312, T122 with 80 kA.	\$1.98	VEPCO	Dominion (100.00%)	6/1/2029
b4000.108	Carson sub 230 kV – replace 40 kA breaker 23872 with 63 kA.	\$0.50	VEPCO	Dominion (100.00%)	6/1/2029
b4000.109	Clifton sub 230 kV – replace 63 kA breakers 201182, SR182, XT2011 with 80 kA.	\$1.98	VEPCO	Dominion (100.00%)	6/1/2029
b4000.110	Doubs sub 500 kV – replace 50 kA breaker DL-59 No. 2CAP with 63 kA.	\$11.50	POTED	APS (100.00%)	6/1/2029
b4000.111	Evergreen Mills sub 230 kV – replace 63 kA breakers H132, H232 with 80 kA.	\$1.32	VEPCO	Dominion (100.00%)	6/1/2029
b4000.112	Goose Creek sub 230 kV – replace 63 kA breaker L1T227 with 80 kA.	\$0.66	VEPCO	Dominion (100.00%)	6/1/2029
b4000.113	Goose Creek sub 500 kV – replace 50 kA breaker SC182 with 63 kA.	\$1.02	VEPCO	Dominion (100.00%)	6/1/2029
b4000.114	Ladysmith S1 sub 230 kV – replace 40 kA breakers 25672, 209072, 256T2090, GT172, GT272, GT372, GT472, GT572 with 63 kA.	\$4.00	VEPCO	Dominion (100.00%)	6/1/2029
b4000.115	Ladysmith sub 500 kV – replace 40 kA breaker 574T581 with 63 kA.	\$1.02	VEPCO	Dominion (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.116	Liberty sub 230 kV – replace 50 kA breaker SC112 with 63 kA.	\$0.50	VEPCO	Dominion (100.00%)	6/1/2029
b4000.117	Lockridge sub 230 kV – replace 63 kA breakers 218872, H12T2188, 222372, H12T2223 with 80 kA.	\$2.64	VEPCO	Dominion (100.00%)	6/1/2029
b4000.118	Loudoun sub 230 kV – replace 63 kA breakers 209452, L152, L252 with 80 kA.	\$1.98	VEPCO	Dominion (100.00%)	6/1/2029
b4000.119	Loudoun Cap sub 230 kV – replace 50 kA breaker SC352 with 63 kA.	\$0.50	VEPCO	Dominion (100.00%)	6/1/2029
b4000.120	Loudoun sub 500 kV – replace 50 kA breakers 502T535, 569T584, H1T569, H2T502, H2T584, SC152 with 63 kA.	\$6.09	VEPCO	Dominion (100.00%)	6/1/2029
b4000.121	Marsh Run sub 230 kV – replace 50 kA breaker 28002, 29902, 280T2039, 299T2040, 203902, 204002 with 63 kA.	\$3.00	VEPCO	Dominion (100.00%)	6/1/2029
b4000.122	Morrisville sub 230 kV – replace 50 kA breaker L1T2039, L1T2040, L2T2039, L2T2040 with 63 kA.	\$2.00	VEPCO	Dominion (100.00%)	6/1/2029
b4000.123	Morrisville sub 500 kV – replace 50 kA breakers H1T541, H1T594, H2T545, H2T569, SC122 with 63 kA.	\$5.08	VEPCO	Dominion (100.00%)	6/1/2029
b4000.124	Mosby sub 500 kV – replace 50 kA breakers 50272, 54672, 55972, 58472, 59072, 502T546, 559T584, SC172, SV172, SV272, XT590 with 63 kA.	\$11.17	VEPCO	Dominion (100.00%)	6/1/2029
b4000.125	Mt Storm sub 500 kV – replace 40 kA breaker G3T572X with 63 kA.	\$1.02	VEPCO	Dominion (100.00%)	6/1/2029
b4000.126	Nimbus sub 230 kV – replace 63 kA breakers 215282, 225532-5, 225532-6, 226034 with 80 kA.	\$2.64	VEPCO	Dominion (100.00%)	6/1/2029
b4000.127	NIVO 1 sub 230 kV – replace 63 kA breaker 2116T2130 with 80 kA (4-breaker ring bus).	\$0.66	VEPCO	Dominion (100.00%)	6/1/2029
b4000.128	North Anna sub 500 kV – replace 40 kA breakers 57502, G102-1, G102-2, G202, G2T575, XT573 with 63 kA.	\$6.09	VEPCO	Dominion (100.00%)	6/1/2029
b4000.129	Ox sub 230 kV – replace 50 kA and 63 kA breakers 201342, 209742, 206342, SC242 with 80 kA.	\$2.64	VEPCO	Dominion (100.00%)	6/1/2029
b4000.130	Ox sub 500 kV – replace 40 kA breakers 56142, H1T539, H2T539 with 63 kA.	\$3.05	VEPCO	Dominion (100.00%)	6/1/2029
b4000.131	Paragon Park sub 230 kV – replace 63 kA breakers 220632, 220732 with 80 kA.	\$1.32	VEPCO	Dominion (100.00%)	6/1/2029
b4000.132	Pleasantview sub 230 kV – replace 63 kA breakers 203T274, 274T2098 with 80 kA.	\$1.32	VEPCO	Dominion (100.00%)	6/1/2029
b4000.133	Pleasantview sub 500 kV – replace 40 kA breaker H322 with 63 kA.	\$1.02	VEPCO	Dominion (100.00%)	6/1/2029
b4000.134	Remington sub 230 kV – replace 40 kA and 50 kA breakers 211462, GT162, GT262, GT362, GT462, 2077T2086, 208662, H962, H9T299 with 63 kA.	\$4.50	VEPCO	Dominion (100.00%)	6/1/2029
b4000.135	Roundtable sub 230 kV – replace 63 kA breakers 203102, 214902, 221402, 222302, 2031T2223, 2149T2214 with 80 kA.	\$3.96	VEPCO	Dominion (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.136	Vint Hill sub 230 kV – replace 63 kA breakers 2101T2174, 2163T2174, 2101T2163 with 80 kA.	\$1.98	VEPCO	Dominion (100.00%)	6/1/2029
b4000.137	Yardley sub 230 kV – replace 63 kA breakers WT2209, WT2213, XT2209, XT2213 with 80 kA.	\$2.64	VEPCO	Dominion (100.00%)	6/1/2029
b4000.201	Smith Mountain 138 kV upgrade: Replace 795 KCM AAC, 37-Str. 795 KCM AAC, 37-Str. PH A B2S1 B2S2 BS1 BS2.	\$0.40	AEP	AEP (100.00%)	6/1/2029
b4000.202	Reconductor 34 miles of Smith Mountain-Redeye 138 kV line.	\$14.90	AEP	AEP (100.00%)	6/1/2029
b4000.203	Reconductor 34 miles of Redeye-Candler's Mountain 138 kV line.	\$21.00	AEP	AEP (100.00%)	6/1/2029
b4000.204	Reconductor 34 miles of Candler's Mountain-Opossum Creek 138 kV line.	\$4.90	AEP	AEP (100.00%)	6/1/2029
b4000.205	Candler's Mountain 138 kV: Replace 1590 KCM AAC, 61-Str. Replace MOAB "Y" Smith Mtn Line.	\$0.20	AEP	AEP (100.00%)	6/1/2029
b4000.206	Opossum Creek 138 kV: Replace Opossum Creek switch.	\$0.10	AEP	AEP (100.00%)	6/1/2029
b4000.207	Leesville station upgrade 138 kV: Replace 795 KCM AAC, 37-Str. IPS Sch. 40 1272 KCM AAC, 61-Str. 1272 KCM AAC, 61-Str. PH A,B,C ALTA VISTA CB-A BUS DISC ALTA VISTA CB-A LINE DISC Wave trap (1200A) relay thermal limit 1356 amps.	\$0.60	AEP	AEP (100.00%)	6/1/2029
b4000.208	Otter 138 kV station upgrade: Replace 795 KCM AAC, 37-Str.	\$0.10	AEP	AEP (100.00%)	6/1/2029
b4000.209	Reconductor 14.4 miles of Altavista-Otter 138 kV line.	\$1.20	AEP	AEP (100.00%)	6/1/2029
b4000.210	Reconductor 14.4 miles of Otter-Johnson Mountain 138 kV line.	\$8.40	AEP	AEP (100.00%)	6/1/2029
b4000.211	Reconductor 14.4 miles of Johnson Mountain-New London 138 kV line.	\$7.68	AEP	AEP (100.00%)	6/1/2029
b4000.300	Rebuild approximately 1.71 miles of 230 kV line 299 from the Marsh Run substation to the Remington CT substation. New conductor has a summer rating of 1573 MVA.	\$5.35	VEPCO	Dominion (100.00%)	6/1/2029
b4000.301	Reconductor approximately 1.24 miles of 230 kV line 280 from Remington – the Marsh Run CT substation to the Remington substation. New conductor has a summer rating of 1573 MVA.	\$3.89	VEPCO	Dominion (100.00%)	6/1/2029
b4000.302	Upgrade line No. 299 terminal equipment, line leads, and bus at Marsh Run substation to be rated to 4000A.	\$2.53	VEPCO	Dominion (100.00%)	6/1/2029
b4000.303	Upgrade line No. 299 terminal equipment, line leads, and bus at Remington CT substation to be rated to 4000A.	\$2.44	VEPCO	Dominion (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.304	Partial reconductor/partial wreck and rebuild of 230 kV line No. 2161 Gainesville-Wheeler. New conductor has a summer rating of 1573 MVA. (Wheeler-Linton tap segment)	\$17.72	VEPCO	Dominion (100.00%)	6/1/2029
b4000.305	Partial reconductor/partial wreck and rebuild of 230 kV line No. 2161 Gainesville-Wheeler. New conductor has a summer rating of 1573 MVA. (Linton tap-Atlantic segment)	\$3.49	VEPCO	Dominion (100.00%)	6/1/2029
b4000.306	Partial reconductor/partial wreck and rebuild of 230 kV line No. 2161 Gainesville-Wheeler. New conductor has a summer rating of 1573 MVA. (Atlantic-Trident segment)	\$1.72	VEPCO	Dominion (100.00%)	6/1/2029
b4000.307	Partial reconductor/partial wreck and rebuild of 230 kV line No. 2161 Gainesville-Wheeler. New conductor has a summer rating of 1573 MVA. (Trident-Gainesville segment)	\$6.16	VEPCO	Dominion (100.00%)	6/1/2029
b4000.308	Upgrade all line No. 2161 terminal equipment at Gainesville to 4000A. A CCVT will also be replaced due to aging.	\$1.78	VEPCO	Dominion (100.00%)	6/1/2029
b4000.309	Upgrade all line No. 2161 terminal equipment Wheeler substation to 4000A.	\$2.25	VEPCO	Dominion (100.00%)	6/1/2029
b4000.310	Revise relay settings at Trident substation.	\$0.03	VEPCO	Dominion (100.00%)	6/1/2029
b4000.311	Rebuild 230 kV line No. 213 and No. 225 from Thelma-Lakeview. New conductor has a summer rating of 1573 MVA.	\$44.69	VEPCO	Dominion (100.00%)	6/1/2029
b4000.312	At Thelma substation, upgrade line lead, wave traps (213WT and 225WT), circuit breaker leads to 4000A. CB switches 22535, 23235, 23238 and 21335 will also be upgraded to 4000A DEB switches. CCVTs 213P1, 213P2 and 213P3 will be replaced due to aging.	\$2.30	VEPCO	Dominion (100.00%)	6/1/2029
b4000.313	At Lakeview substation, upgrade wave traps 213WT and 225WT, line leads and circuit breaker leads to 4000A. Upgrade CB switches 22565 and 22564 to 4000A double-end break switches. Replace CCVTs 225P1, 225P2 and 225P3 due to aging.	\$1.72	VEPCO	Dominion (100.00%)	6/1/2029
b4000.314	Reconductor 230 kV line No. 2003 Chesterfield-Tyler segment. New conductor has a summer rating of 1573 MVA.	\$3.16	VEPCO	Dominion (100.00%)	6/1/2029
b4000.315	Reconductor 230 kV line No. 2003 Tyler-Poe segment. New conductor has a summer rating of 1573 MVA.	\$20.61	VEPCO	Dominion (100.00%)	6/1/2029
b4000.316	At Poe substation, uprate all line No. 2003 terminal equipment, line leads and bus to be rated to 4000A.	\$1.79	VEPCO	Dominion (100.00%)	6/1/2029
b4000.317	At Tyler substation, upgrade the necessary line terminal equipment to maintain 4000A at Tyler substation.	\$0.54	VEPCO	Dominion (100.00%)	6/1/2029
b4000.318	Revise relay settings at Chesterfield substation.	\$0.02	VEPCO	Dominion (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.319	Reconductor 230 kV line No. 2002 Carson-Poe. New conductor has a summer rating of 1573 MVA.	\$30.29	VEPCO	Dominion (100.00%)	6/1/2029
b4000.320	At Carson substation, upgrade all line No. 2002 terminal equipment at Carson to 4000A. CCVTs will also be replaced due to aging.	\$0.89	VEPCO	Dominion (100.00%)	6/1/2029
b4000.321	At Poe substation, upgrade all line No. 2002 terminal equipment at Carson to 4000A. CCVTs will also be replaced due to aging.	\$0.81	VEPCO	Dominion (100.00%)	6/1/2029
b4000.322	Build a new 230 kV line from Nokesville-Hornbaker using the vacant arms of the double circuit monopole structures installed as part of previous project 993027. New conductor has a summer rating of 1573 MVA.	\$15.28	VEPCO	Dominion (100.00%)	6/1/2029
b4000.323	Upgrade terminal equipment at Nokesville substation. The project adds one more line to Nokesville, including the installation of one 230 kV breaker and two 230 kV switches.	\$1.43	VEPCO	Dominion (100.00%)	6/1/2029
b4000.324	Upgrade terminal equipment at Hornbaker substation. This project provides for installing a new 230 kV 4000A rated line terminal at Hornbaker to accommodate the new line to Nokesville.	\$1.43	VEPCO	Dominion (100.00%)	6/1/2029
b4000.325	Build a new 26.38-mile 230 kV line from Elmont-Ladysmith on the existing 5-2 structures between the two stations. New conductor has a summer rating of 1573 MVA.	\$26.09	VEPCO	Dominion (100.00%)	6/1/2029
b4000.326	At Elmont substation, install/upgrade associated equipment to accommodate a 4000A line rating for the new 230 kV line between Elmont-Ladysmith.	\$2.19	VEPCO	Dominion (100.00%)	6/1/2029
b4000.327	Upgrade/install equipment at Ladysmith substation to 4000A. Expansion will be required to accommodate a total of three (3) new 230 kV strings of breaker and a half scheme.	\$8.72	VEPCO	Dominion (100.00%)	6/1/2029
b4000.328	Construct a new 24.5-mile 230 kV line from Cloverhill substation to Ox substation.	\$287.54	VEPCO	Dominion (100.00%)	6/1/2029
b4000.329	At Ox substation, install the necessary associated equipment to accommodate the new line between Cloverhill-Ox. This project also includes expanding the substation with associated security level 1 fencing and super post structure needed.	\$6.37	VEPCO	Dominion (100.00%)	6/1/2029
b4000.330	At Cloverhill substation, install the necessary associated equipment to accommodate the new line between Cloverhill-Ox. This project also includes demolishing and reconstructing the existing bus system and roadway.	\$2.27	VEPCO	Dominion (100.00%)	6/1/2029
b4000.331	Construct a new 230 kV single circuit line from Raines substation to Cloud substation to solve electrical violations cause by the significant load growth in South Hill, Virginia. The scope also includes an idle 230 kV circuit being installed between these stations.	\$77.50	VEPCO	Dominion (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.332	At Cloud substation, upgrade substation terminal equipment to 4000A.	\$1.43	VEPCO	Dominion (100.00%)	6/1/2029
b4000.333	At Raines substation, upgrade substation terminal equipment to 4000A.	\$1.32	VEPCO	Dominion (100.00%)	6/1/2029
b4000.334	Reconductor 115 kV line No. 121 from Poe-Prince George. Specifically, line No. 121 will be reconducted and converted to 230 kV from Poe substation to Prince George substation.	\$14.62	VEPCO	Dominion (100.00%)	6/1/2029
b4000.335	At Poe substation, install a new 230 kV six-breaker ultimate ring bus which will fit the station to current 230 kV standards. The substation scope includes the installation of 230 kV breaker and half GIS bus. Work at Poe substation is associated with line No. 121 reconductor.	\$38.28	VEPCO	Dominion (100.00%)	6/1/2029
b4000.336	Build a new 230-115 kV Prince George substation along the existing 115 or 230 kV corridor. The substation scope includes the installation of 230 kV breakers and 1-115 kV breaker along with its associated terminal equipment initially but will have provision for making it a six-breaker ring (both 230 and 115 kV) in future. The existing 230-115 kV transformer at Prince George will be relocated to serve this new substation.	\$26.62	VEPCO	Dominion (100.00%)	6/1/2029
b4000.337	Extend a new 230 kV line approximately 7.85 miles between the existing Morrisville and Anderson Branch substations. The existing tower structures currently supporting the Bristers to Morrisville 500 kV Line No. 545 will be used to support this new line as shared tower structures.	\$5.26	VEPCO	Dominion (100.00%)	6/1/2029
b4000.338	At Morrisville substation, install/upgrade substation terminal equipment to 4000A.	\$0.95	VEPCO	Dominion (100.00%)	6/1/2029
b4000.339	At Anderson Branch substation, install/upgrade substation terminal equipment to 4000A.	\$0.95	VEPCO	Dominion (100.00%)	6/1/2029
b4000.340	Uprate existing Goose Creek 500/230 kV transformer to 1440 MVA.	\$34.01	VEPCO	Dominion (100.00%)	6/1/2029
b4000.348	Build a new 500/230 kV substation called Kraken. The 500 kV, 5000A ring bus will be set up for a redundant breaker configuration. Install (2) 1400MVA 500/230 kV transformers. A new redundant breaker ring will be added at Kraken to accommodate the new 500 kV line from North Anna to Kraken.	\$84.41	VLT	Dominion (100.00%)	6/1/2029
b4000.358	Build a new 765/500/230 kV substation called Yeat. Install (1) 500/230 kV transformer. Cut in 230 kV line Vint Hill-Elk Run into Yeat substation.	\$45.60	VLT	Dominion (100.00%)	6/1/2029

Attachment B – Reliability Project Multi-Zone Allocations

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b3907.1	Reconductor entire 2.5 miles of North Philadelphia to Master 230 kV line.	\$6.49	PECO	AEC (0.76%)/JCPL (1.45%)/PECO (94.22%)/PSEG (3.29%)/RE (0.13%)/Neptune (0.15%)	6/1/2029
b3907.2	Richmond to Waneeta 230 kV line: Rebuild entire 0.95 miles of existing UGT and rebuild entire 2.23 miles of existing OHT.	\$29.40	PECO	AEC (9.99%)/JCPL (8.34%)/PECO (55.62%)/PSEG (24.31%)/RE (0.94%)/Neptune (0.80%)	6/1/2028
b3907.3	Reconductor 2.12 miles of North Philadelphia to Waneeta 230 kV line.	\$7.33	PECO	AEC (10.75%)/JCPL (14.26%)/PECO (38.58%)/PSEG (33.62%)/RE (1.31%)/Neptune (1.48%)	6/1/2028
b3909.1	Juniata 500 kV yard expansion/reconfiguration to include one new bay and eliminate the line fault stuck breaker.	\$22.20	PPL	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: PPL (100.00%)	6/1/2028
b3915.1	Reconductor 1.5 miles of 345 kV lines 1202 and 1227 from Dresden to Mulberry with two conductor bundled 1033.5 ACSS conductor. Modify and replace towers as necessary to accommodate the higher mechanical loads of the bundled conductor.	\$12.33	ComEd	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: ComEd (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b3915.2	Install 345 kV circuit breaker on line 1202 (Dresden-Mulberry 345 kV) and upgrade disconnects and associated equipment at Dresden 345 kV substation.	\$3.77	ComEd	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: ComEd (100.00%)	6/1/2029
b3915.3	Upgrade disconnects and associated equipment at Mulberry 345 kV substation.	\$0.17	ComEd	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: ComEd (100.00%)	6/1/2029
b3919.1	Reconfigure Maliszewski 765 kV station from two breakers to a six-breaker ring bus. Install a new 765/345 kV transformer. Establish new 345 kV breaker yard with three-string breaker and a half to include a line exit to Hyatt and a line exit to Corridor. Loop the existing Hyatt-West Millersport 345 kV line into the new established 345 kV yard at the Maliszewski station.	\$79.74	AEP	AEP (85.10%)/Dayton (9.33%)/DEOK (5.48%)/OVEC (0.09%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b3929.1	Rebuild approximately 33.09 miles of 500 kV line No. 579 from structure 579/1 inside Septa substation to structure 579/193 inside Yadkin substation.	\$191.25	VEPCO	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: Dominion (100.00%)	6/1/2029
b3929.2	At Septa substation, upgrade CB (579T586), breaker switches (56288, 57985, 58688 and 57988) and line leads to 5000A rating to support line No. 519 rebuild.	\$3.84	VEPCO	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: Dominion (100.00%)	6/1/2029
b3929.3	At Yadkin substation, upgrade line leads to 5000A rating to support line No. 579 rebuild.	\$0.44	VEPCO	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: Dominion (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.1	Add one 765 kV breaker at Amos substation to expand the breaker-and-a-half scheme to accommodate the new Amos-Welton Spring 765 kV line.	\$30.87	AEP	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (14.67%)/BGE (8.11%)/Dominion (66.09%)/DPL (2.15%)/PEPCO (8.98%)	6/1/2029
b4000.2	Construct a new 175 miles Amos to Welton Spring 765 kV line (AEP section of the line is 30 miles).	\$150.00	VLT	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (14.67%)/BGE (8.11%)/Dominion (66.09%)/DPL (2.15%)/PEPCO (8.98%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.3	Construct a new 175-mile Amos to Welton Spring 765 kV line (APS/FE section of the line is 145 miles).	\$725.00	VLT	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (14.67%)/BGE (8.11%)/Dominion (66.09%)/DPL (2.15%)/PEPCO (8.98%)	6/1/2029
b4000.4	Construct a new 765 kV switchyard (Welton Spring).	\$92.28	VLT	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (14.67%)/BGE (8.11%)/Dominion (66.09%)/DPL (2.15%)/PEPCO (8.98%)	6/1/2029
b4000.5	Install four 250 MVAR shunt capacitors, and a +/-500 MVAR STATCOM at Welton Spring 765 kV substation.	\$121.54	VLT	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.6	Construct a new 86-mile Welton Spring to Rocky Point 765 kV line (Welton Spring to DVP local area, roughly 70.7 miles).	\$353.50	VLT	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (14.67%)/BGE (8.11%)/Dominion (66.09%)/DPL (2.15%)/PEPCO (8.98%)	6/1/2029
b4000.7	Construct a new 86-mile Welton Spring to Rocky Point 765 kV line (in DVP local area/Millville to Lovettsville section, roughly 10.5 miles).	\$52.50	VLT	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (14.67%)/BGE (8.11%)/Dominion (66.09%)/DPL (2.15%)/PEPCO (8.98%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.8	Construct a new 86-mile Welton Spring to Rocky Point 765 kV line (DVP local area to Rocky Point, roughly 4.8 miles).	\$24.00	VLT	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (14.67%)/BGE (8.11%)/Dominion (66.09%)/DPL (2.15%)/PEPCO (8.98%)	6/1/2029
b4000.9	Construct a new substation called Rocky Point with a 765 kV and a 500 kV yard. Loop in the Doubs-Goose Creek 500 kV line, the Doubs-Aspen 500 kV line, and the Woodside-Goose Creek 500 kV line. Install two 765/500 kV transformer at Rocky Point substation.	\$254.70	VLT	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (14.99%)/BGE (8.29%)/Dominion (67.55%)/PEPCO (9.17%)	6/1/2029
b4000.10	Install four 250 MVAR cap banks (two on 765 and two on 500 kV side) and a +/-500 MVAR STATCOM on the 765 kV Rocky Point yard.	\$121.00	VLT	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.11	Expand Black Oak substation to accommodate the connection of the 502 Jct-Woodside 500 kV line and loop the 502 Jct-Woodside 500 kV line into the Black Oak substation by constructing ~0.85 miles of new 500 kV line into and out of Black Oak 500 kV substation.	\$19.23	POTED	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (8.93%)/BGE (13.63%)/Dominion (61.66%)/PEPCO (15.78%)	6/1/2029
b4000.12	Upgrade the terminal equipment on the Doubs No. 1 500/230 kV transformer.	\$0.43	POTED	APS (74.10%)/PEPCO (25.90%)	6/1/2029
b4000.13	Terminate the Woodside-Goose Creek 500 kV line into Doubs substation, creating the Woodside-Doubs No. 2 500 kV line. Remove the Chanceford-Doubs and Rocky Point-Doubs line terminations at the Doubs substation and connect the two lines through a 0.6-mile 500 kV bypass line around the Doubs substation.	\$13.97	POTED	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: AEC (0.20%)/APS (0.06%)/BGE (0.71%)/Dominion (89.06%)/DPL (0.36%)/JCPL (0.33%)/ME (0.15%)/PECO (0.73%)/PEPCO (7.93%)/PSEG (0.42%)/RE (0.02%)/Neptune (0.03%)	6/1/2032

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.200	Broadford 765 kV upgrade: Replace Jackson's Ferry CB Q2.	\$4.50	AEP	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: AEP (21.60%)/APS (12.36%)/BGE (8.28%)/Dominion (46.81%)/PEPCO (10.95%)	6/1/2029
b4000.251	Replace the wave trap and upgrade the relay at Cloverdale 765 kV substation.	\$1.00	AEP	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: AEP (3.00%)/APS (8.96%)/BGE (6.53%)/Dominion (72.75%)/PEPCO (8.76%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.252	Replace the wave trap and upgrade the relay at Joshua Falls 765 kV substation.	\$1.00	AEP	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: AEP (3.00%)/APS (8.96%)/BGE (6.53%)/Dominion (72.75%)/PEPCO (8.76%)	6/1/2029
b4000.341	Remove the 500 kV conductor previously planned to terminate into the Vint Hill 500 kV substation and extend approximately 0.2 miles of conductor to fly-over the site.	\$0.00	VEPCO	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: Dominion (100.00%)	6/1/2029
b4000.342	Remove the terminal equipment and substation work required for the termination of the Morrisville-Wishing Star 500 kV line into Vint Hill.	\$0.00	VEPCO	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: Dominion (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.343	Uprate bus at Brambleton to support 500 kV line No. 558 (Aspen to Brambleton) uprate.	\$0.00	VEPCO	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: Dominion (100.00%)	6/1/2029
b4000.344	Build a 500 kV line from North Anna substation (bypassing Ladysmith Substation) to a new substation called Kraken. New conductor to have a minimum summer normal rating of 4357 MVA.	\$186.67	VEPCO	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: Dominion (91.69%)/PEPCO (8.31%)	6/1/2029
b4000.345	Build a 500 kV line from a new substation called Kraken to a new substation called Yeat. New conductor to have a minimum summer normal rating of 4357 MVA.	\$394.71	VEPCO	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: Dominion (100.00%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.346	Cut-in 500 kV line from Kraken substation into Yeat substation.	\$5.00	VLT	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: Dominion (100.00%)	6/1/2029
b4000.347	Upgrade/install equipment at North Anna substation to 5000A to support the new conductor rating.	\$9.12	VEPCO	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: Dominion (91.69%)/PEPCO (8.31%)	6/1/2029
b4000.349	Update relay settings at Ladysmith to change the destination of 500 kV line No. 568 from Possum Point to Kraken.	\$0.04	VEPCO	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: BGE (13.28%)/Dominion (64.48%)/PEPCO (22.24%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.350	Update relay settings at Possum Point to change the destination of 500 kV line No. 568 from Ladysmith to Kraken.	\$0.04	VEPCO	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: BGE (8.30%)/Dominion (78.64%)/PEPCO (13.06%)	6/1/2029
b4000.351	Cut in line No. 568 Ladysmith-Possum Point into Kraken, creating line No. 568 Ladysmith to Kraken.	\$3.60	VEPCO	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: BGE (8.30%)/Dominion (78.64%)/PEPCO (13.06%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.352	Cut in line No. 568 Ladysmith-Possum Point into Kraken, creating new line from Kraken to Possum Point.	\$3.60	VEPCO	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: BGE (13.28%)/Dominion (64.48%)/PEPCO (22.24%)	6/1/2029
b4000.353	Upgrade 500 kV terminal equipment at Elmont substation.	\$8.90	VEPCO	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (9.79%)/BGE (6.14%)/Dominion (75.61%)/PEPCO (8.46%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.354	Expand Ladysmith substation to add redundant circuit breakers to the middle breakers on both 500 kV strings (574T575 and 568T581). The equipment including switches 57518, 57515, H115 will be replaced with 5000A equipment.	\$8.24	VEPCO	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (9.79%)/BGE (6.14%)/Dominion (75.61%)/PEPCO (8.46%)	6/1/2029
b4000.355	Build a new 156-mile 765 kV line from Joshua Falls-Yeat (Roughly 69.3 miles in AEP section).	\$341.17	VLT	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (9.11%)/BGE (6.49%)/Dominion (75.72%)/PEPCO (8.68%)	6/1/2029

Upgrade ID	Description	Cost Estimate (\$M)	TO	Cost Responsibility	Required IS Date
b4000.356	Build a new 156-mile 765 kV line from Joshua Falls-Yeat (Roughly 86.7 miles in Dominion section).	\$426.83	VLT	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (9.11%)/BGE (6.49%)/Dominion (75.72%)/PEPCO (8.68%)	6/1/2029
b4000.357	Build a new 765/500/230 kV substation called Yeat. Install (2) 765/500 kV transformers. Cut in 500 kV line Bristers-Ox and 500 kV line Meadowbrook-Vint Hill into Yeat.	\$197.10	VLT	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: Dominion (89.71%)/PEPCO (10.29%)	6/1/2029
b4000.359	Add (2) 765 kV breakers at Joshua Falls substation. Substation expansion is required to add the additional breakers.	\$99.19	AEP	Load-Ratio Share Allocation: AEC (1.58%)/AEP (13.71%)/APS (5.49%)/ATSI (7.69%)/BGE (4.16%)/ComEd (13.25%)/Dayton (2.07%)/DEOK (3.18%)/Dominion (14.20%)/DPL (2.57%)/DL (1.65%)/EKPC (2.30%)/JCPL (3.80%)/ME (1.88%)/OVEC (0.06%)/PECO (5.32%)/PENELEC (1.81%)/PEPCO (3.79%)/PPL (4.58%)/PSEG (6.24%)/RE (0.25%)/Neptune (0.42%) DFAX Allocation: APS (9.11%)/BGE (6.49%)/Dominion (75.72%)/PEPCO (8.68%)	6/1/2029

Attachment C – Interconnection Queue Projects With System Impact Study Reports Issued

Generation Interconnection Requests

	Transmission Owner	Fuel Type	MW Capacity (FTIR/FTWR)	MW Energy (nFTIR/nFTWR)
AE1-070	ComEd	Natural Gas	135	0
AE1-092	Dayton	Solar	96.4	206.6
AE1-114	ComEd	Wind	34	150
AE1-144	EKPC	Solar	66.8	100
AE1-148	Dominion	Solar	54	90
AE1-172	ComEd	Wind	44.88	255
AE2-038	EKPC	Solar	53.5	80
AE2-053	Dominion	Storage	20	20
AE2-107	ComEd	Solar	54	90
AE2-123	Dominion	Offshore Wind	163.3	800.1
AE2-124	Dominion	Offshore Wind	148.5	800.1
AE2-130	AEP	Solar/Storage	480	800
AE2-138	EKPC	Solar	20.8	260
AE2-147	Dominion	Solar	90	150
AE2-156	Dominion	Storage	100	100
AE2-173	ComEd	Storage	50	50
AE2-185	Dominion	Solar	36	60
AE2-187	Dominion	Solar	36	60
AE2-210	EKPC	Solar	7.2	90
AE2-223	ComEd	Wind	19.5	150
AE2-250	Dominion	Solar	54	82.5
AE2-261	ComEd	Solar	179.4	299
AE2-270	Dominion	Solar/Storage	150	150
AE2-275	EKPC	Solar	7.2	90
AE2-283	Dominion	Solar	28	53
AE2-291	Dominion	Solar	61.2	102
AE2-292	Dominion	Solar	76.2	127
AE2-308	EKPC	Solar	60	100
AE2-313	Dominion	Solar	188.4	314
AE2-321	ComEd	Solar	67	100
AE2-325	AEP	Storage	31.32	52.2
AE2-341	ComEd	Solar	100.6	150
AF1-030	ComEd	Solar	66.9	100
AF1-048	ComEd	Storage	31.32	52.2

	Transmission Owner	Fuel Type	MW Capacity (FTIR/FTWR)	MW Energy (nFTIR/nFTWR)
AF1-088	AEP	Not Specified	1000	1000
AF1-099	APS	Solar/Storage	75.9	126.5
AF1-100	PENELEC	Solar/Storage	150	250
AF1-123	Dominion	Offshore Wind	253.2	833
AF1-124	Dominion	Offshore Wind	254.1	836
AF1-125	Dominion	Offshore Wind	249.3	820
AF1-128	Dominion	Natural Gas	569	569
AF1-142	APS	Solar/Storage	121.44	202.4
AF1-152	Dominion	Solar	30	50
AF1-161	AEP	Storage	25	50
AF1-176	AEP	Solar/Storage	155.684	300
AF1-200	ComEd	Not Specified	2035	2100
AF1-201	Dominion	Solar/Storage	90	150
AF1-204	AEP	Wind	63.75	255
AF1-221	AEP	Solar	25.2	180
AF1-233	EKPC	Solar	113.1	188.5
AF1-238	AEC	Storage	20	50
AF1-240	PENELEC	Solar	11	15.75
AF1-252	ComEd	Storage	62.4	62.4
AF1-253	ComEd	Storage	43.2	43.2
AF1-256	EKPC	Solar	48	80
AF1-266	Dominion	Storage	74.5	74.5
AF1-276	PENELEC	Solar	67	100
AF1-277	PENELEC	Solar	67	100
AF1-278	PENELEC	Solar	53.44	80
AF1-280	ComEd	Solar	0	200
AF1-281	ComEd	Storage	3	20
AF1-293	Dominion	Solar/Storage	108.37	127.86
AF1-294	Dominion	Solar	22.2	41
AF1-296	ComEd	Wind	37.3296	212.1
AF1-301	Dominion	Solar	70	117
AF1-306	APS	Storage	49.3	197.2
AF1-314	ComEd	Wind	37.2	212
AF1-318	ComEd	Wind	52.7947	299.97
AF1-323	AEP	Solar	33	55
AF2-008	AEP	Not Specified	500	1000
AF2-010	PENELEC	Solar	46	77
AF2-016	AEC	Storage	96	240

	Transmission Owner	Fuel Type	MW Capacity (FTIR/FTWR)	MW Energy (nFTIR/nFTWR)
AF2-021	AEC	Storage	8	20
AF2-025	AEC	Storage	8	20
AF2-034	ComEd	Solar/Storage	10.8	20
AF2-035	Dominion	Solar	48	72
AF2-037	Dominion	Solar	56.4	94
AF2-041	ComEd	Solar	180	300
AF2-042	Dominion	Solar	300	500
AF2-046	Dominion	Solar	93	140
AF2-047	Dominion	Solar	99.8	150
AF2-050	PENELEC	Solar	30	50
AF2-056	PPL	Solar	33	50
AF2-063	Dominion	Solar	90	150
AF2-065	Dominion	Solar	76.5	150
AF2-068	AEP	Solar	90	150
AF2-069	ComEd	Wind	2.2	9.3
AF2-070	ComEd	Wind	3.1	18.15
AF2-080	Dominion	Solar	48.5	70
AF2-081	Dominion	Solar	56	80
AF2-095	ComEd	Solar	96	160
AF2-096	ComEd	Solar	180	300
AF2-111	EKPC	Solar	150	250
AF2-115	Dominion	Solar	13.5	25
AF2-120	Dominion	Solar	37.2	62
AF2-121	PENELEC	Solar	12	20
AF2-126	ATSI	Solar	8	12
AF2-128	ComEd	Wind	13.992	79.5
AF2-142	ComEd	Solar	90	150
AF2-143	ComEd	Solar	90	150
AF2-164	PENELEC	Solar	48	80
AF2-171	Dominion	Solar	90	150
AF2-173	AEP	Solar	84	140
AF2-176	ATSI	Solar	36	60
AF2-177	AEP	Wind	26	200
AF2-182	ComEd	Solar	0	300
AF2-183	ComEd	Storage	32	80
AF2-193	DPL	Offshore Wind	119	440
AF2-194	DPL	Offshore Wind	119	440
AF2-199	ComEd	Solar	60	100

	Transmission Owner	Fuel Type	MW Capacity (FTIR/FTWR)	MW Energy (nFTIR/nFTWR)
AF2-200	ComEd	Solar	120	200
AF2-201	ComEd	Wind/Solar	51.4	199
AF2-209	AEP	Solar	19.6	140
AF2-222	Dominion	Solar	100	167
AF2-225	ComEd	Solar	63	150
AF2-226	ComEd	Storage	20	50
AF2-263	Dayton	Solar	58.8	98
AF2-296	PENELEC	Solar	12	20
AF2-297	Dominion	Solar	48	80
AF2-299	Dominion	Solar	10.8	18
AF2-302	AEP	Solar	12	20
AF2-307	EKPC	Solar	39.6	64.2
AF2-319	ComEd	Storage	20	50
AF2-321	ATSI	Solar	88.8	148
AF2-324	Dominion	Solar	45	75
AF2-329	ComEd	Storage	42.2	42.2
AF2-335	AEP	Solar	60	100
AF2-348	EKPC	Solar	150	250
AF2-349	ComEd	Solar	0	300
AF2-350	ComEd	Solar	60	100
AF2-351	ComEd	Storage	8	20
AF2-355	EKPC	Solar	135	225
AF2-358	DPL	Solar	60	100
AF2-370	AEP	Storage	20	0
AF2-381	PENELEC	Solar	40.2	60
AF2-388	AEP	Wind	35.2	200
AF2-392	ComEd	Wind	35.2	199
AF2-393	ComEd	Storage	55	60
AF2-394	ComEd	Storage	40	40
AF2-396	AEP	Solar/Storage	200	200
AF2-397	Dominion	Solar/Storage	78	130
AF2-403	Dominion	Storage	8	0
AF2-404	Dominion	Storage	0	0
AF2-406	PENELEC	Storage	75	75
AF2-407	AEP	Storage	300	300
AF2-412	PENELEC	Storage	100	100
AF2-441	ComEd	Storage	80	200
AG1-005	ComEd	Solar	120	200

	Transmission Owner	Fuel Type	MW Capacity (FTIR/FTWR)	MW Energy (nFTIR/nFTWR)
AG1-008	Dominion	Solar	93	140
AG1-011	Dominion	Storage	160	160
AG1-019	Dominion	Storage	100	100
AG1-021	Dominion	Solar	10.8	20
AG1-022	AEP	Solar	12	20
AG1-030	Dominion	Solar	90	150
AG1-051	ATSI	Storage	20	20
AG1-060	PPL	Solar	25.7	174.8
AG1-061	DPL	Solar	65.58	109.3
AG1-075	Dominion	Solar	93.2	150
AG1-082	Dominion	Solar/Storage	12	20
AG1-083	Dominion	Solar/Storage	12	20
AG1-090	PENELEC	Solar/Storage	30	95
AG1-098	Dominion	Solar	64.2	107
AG1-105	Dominion	Solar	54	90
AG1-106	Dominion	Solar	16	23
AG1-109	AEP	Storage	25	0
AG1-118	ComEd	Solar	180	300
AG1-119	ComEd	Solar	180	300
AG1-120	ComEd	Solar	71.16	118.6
AG1-121	ComEd	Wind	51.9	294.7
AG1-123	AEP	Solar	38.5	65
AG1-124	AEP	Solar	53.01	90
AG1-127	ComEd	Solar	57.1	95.1
AG1-133	Dominion	Solar	76.8	128
AG1-134	Dominion	Solar	60	100
AG1-135	Dominion	Solar	36	60
AG1-146	Dominion	Solar	18	30
AG1-147	Dominion	Solar	42	70
AG1-151	Dominion	Storage	100	100
AG1-153	Dominion	Storage	30	75
AG1-154	Dominion	Storage	20	50
AG1-160	Dominion	Storage	30	75
AG1-166	Dominion	Solar	6	20
AG1-167	Dominion	Solar	4.5	15
AG1-168	Dominion	Solar	4.5	15
AG1-169	Dominion	Solar	4.5	15
AG1-170	Dominion	Solar	3	10

	Transmission Owner	Fuel Type	MW Capacity (FTIR/FTWR)	MW Energy (nFTIR/nFTWR)
AG1-174	Dominion	Solar	12	20
AG1-178	Dominion	Solar	324.94	500
AG1-179	Dominion	Solar	33.66	50
AG1-180	Dominion	Storage	16.4	0
AG1-181	Dominion	Solar	101.09	150
AG1-184	Dominion	Storage	175.06	0
AG1-185	Dominion	Storage	48.91	0
AG1-194	AEP	Solar/Storage	27	79
AG1-195	Dominion	Solar/Storage	150	150
AG1-196	Dominion	Solar/Storage	150	150
AG1-199	ATSI	Storage	300	300
AG1-219	AEP	Solar/Storage	39.5	75
AG1-224	AEP	Solar	126	210
AG1-225	AEP	Solar	207	346
AG1-226	AEP	Solar	142	450
AG1-236	ComEd	Wind	23.4	180
AG1-257	APS	Solar	8.3	19.9
AG1-272	Dominion	Solar/Storage	6	20
AG1-273	Dominion	Solar/Storage	6	20
AG1-274	Dominion	Solar/Storage	6	20
AG1-277	PPL	Solar/Storage	32	40
AG1-278	PPL	Solar/Storage	15.42	20
AG1-279	PPL	Solar/Storage	15.42	20
AG1-284	Dominion	Solar	42	70
AG1-285	Dominion	Solar	75	125
AG1-288	Dominion	Solar	96.59	150
AG1-289	Dominion	Storage	53.41	0
AG1-297	AEP	Storage	75	300
AG1-298	ComEd	Storage	500	500
AG1-305	Dominion	Solar	37.8	63
AG1-306	EKPC	Solar	39	65
AG1-308	PENELEC	Solar	8.3	19.9
AG1-319	AMPT	Solar	39	65
AG1-321	ComEd	Solar	126.4	200
AG1-323	Dayton	Solar/Storage	40	40
AG1-342	Dominion	Solar	21.6	36
AG1-343	Dominion	Solar/Storage	35	62.5
AG1-346	Dominion	Solar	12	20

	Transmission Owner	Fuel Type	MW Capacity (FTIR/FTWR)	MW Energy (nFTIR/nFTWR)
AG1-353	EKPC	Solar	58.8	98
AG1-354	EKPC	Solar	90	150
AG1-359	Dominion	Storage	127.5	127.5
AG1-364	Dominion	Storage	40	100
AG1-367	AEP	Solar	60	100
AG1-371	ComEd	Storage	30	0
AG1-373	ComEd	Storage	10	0
AG1-374	ComEd	Solar	180	300
AG1-375	AEP	Solar	60	100
AG1-377	PENELEC	Solar	6	20
AG1-378	PENELEC	Solar	6	20
AG1-379	ComEd	Solar	137.2	200
AG1-384	Dominion	Solar/Storage	6	20
AG1-389	-	-	-	-
AG1-390	-	-	-	-
AG1-391	-	-	-	-
AG1-392	PENELEC	Solar	18	30
AG1-393	Dominion	Solar	12	20
AG1-394	Dominion	Solar	5	14.2
AG1-399	ComEd	Wind	52.8	300
AG1-400	ComEd	Storage	150	150
AG1-402	ComEd	Storage	150	150
AG1-404	ComEd	Storage	100	100
AG1-410	AEP	Solar	180	300
AG1-411	AEP	Storage	100	100
AG1-412	Dominion	Storage	80	200
AG1-421	Dominion	Solar	120	200
AG1-422	Dominion	Storage	50	50
AG1-423	ComEd	Solar	120	200
AG1-424	AEP	Solar	60	100
AG1-425	ATSI	Storage	20	0
AG1-426	AEP	Storage	17	0
AG1-427	Dominion	Solar	71.136	118.56
AG1-428	Dominion	Solar	28.08	46.8
AG1-432	Dominion	Solar	120	200
AG1-433	AEP	Wind	17.6	100
AG1-434	ComEd	Wind	35.2	175
AG1-435	ComEd	Wind	35.2	150

	Transmission Owner	Fuel Type	MW Capacity (FTIR/FTWR)	MW Energy (nFTIR/nFTWR)
AG1-436	AEP	Solar	75	125
AG1-437	Dominion	Solar	60	100
AG1-438	Dominion	Solar	60	100
AG1-439	Dominion	Solar	150	250
AG1-440	Dominion	Solar	45	75
AG1-441	Dominion	Solar	45	75
AG1-442	Dominion	Storage	35	35
AG1-443	Dominion	Storage	35	35
AG1-444	Dominion	Storage	80	80
AG1-445	Dominion	Storage	26	26
AG1-446	Dominion	Storage	26	26
AG1-447	AEP	Storage	55	55
AG1-449	Dominion	Storage	122	122
AG1-450	DPL	Storage	25	25
AG1-451	Dominion	Storage	20	20
AG1-453	AEP	Solar	84	140
AG1-454	AEP	Storage	50	50
AG1-456	AEP	Storage	30	30
AG1-460	ComEd	Storage	12	30
AG1-462	ComEd	Solar	153	255
AG1-471	EKPC	Solar	32.4	54
AG1-473	PENELEC	Solar/Storage	32.34	53.9
AG1-483	PEPCO	Storage	500	500
AG1-488	EKPC	Solar	42	70
AG1-494	AEP	Storage	20	50
AG1-497	DPL	Solar/Storage	84	120
AG1-502	Dominion	Solar/Storage	64	80
AG1-503	Dominion	Solar/Storage	16	20
AG1-505	Dominion	Solar	90	150
AG1-506	Dominion	Solar	90	150
AG1-522	AEP	Solar	180	300
AG1-523	AEP	Solar	180	300
AG1-524	AEP	Solar	180	300
AG1-525	AEP	Solar	180	300
AG1-526	EKPC	Solar	133.2	222
AG1-529	DPL	Solar	48.9	75
AG1-531	Dominion	Solar	32.4	54
AG1-535	ComEd	Storage	85	85

	Transmission Owner	Fuel Type	MW Capacity (FTIR/FTWR)	MW Energy (nFTIR/nFTWR)
AG1-536	Dominion	Storage	32	75
AG1-537	Dominion	Storage	32	75
AG1-538	Dominion	Storage	32	75
AG1-539	Dominion	Storage	32	75
AG1-547	Dominion	Solar/Storage	32.54	50
AG1-548	PENELEC	Solar/Storage	45	150
AG1-549	PENELEC	Solar/Storage	85	100
AG1-551	Dominion	Solar	13.1	19.99
AG1-552	Dominion	Solar	12.2	18
AG1-553	ComEd	Solar	0	284
AG1-556	Dominion	Storage	100	100
AD2-134	ComEd	Wind	21.2	105.9
AE1-166	ComEd	Solar	78	150
AE1-205	ComEd	Solar	84	200
AE1-227	AEP	Solar	30.69	49.5
AE2-045	AEP	Solar	84.28	200
AE2-048	AEP	Solar	147.7	220
AE2-051	Dominion	Solar	90	150
AE2-136	AEP	Solar	83.2	114
AE2-139	PENELEC	Solar	60.3	100.5
AE2-152	ComEd	Solar	90	150
AE2-220	AEP	Solar	52.5	125
AE2-234	AEP	Solar	24.1	35
AE2-259	Dominion	Solar	60	100
AE2-298	AEP	Solar	29.9	49.9
AE2-326	AEP	Storage	31.32	52.2
AE2-339	EKPC	Solar	26.8	40
AF1-049	AEP	Solar	75	125
AF1-084	AEP	Solar	54.1	85
AF1-091	AEP	Solar	138.4	180
AF1-112	APS	Solar	8.9	15
AF1-129	Dominion	Natural Gas	569	569
AF1-141	AEP	Solar	62.8	96
AF1-143	PENELEC	Solar	60	100
AF1-144	AEP	Solar	66.9	100
AF1-170	PENELEC	Solar	46.5	77.5
AF1-206	ATSI	Solar	119.4	199
AF1-208	AEC	Solar/Storage	27	45

	Transmission Owner	Fuel Type	MW Capacity (FTIR/FTWR)	MW Energy (nFTIR/nFTWR)
AF1-211	ME	Solar	9	15
AF1-212	ME	Solar	9	15
AF1-231	DPL	Storage	7.6	19
AF1-237	0	Storage	80	170
AF1-239	AEC	Storage	12	30
AF1-245	PSEG	Storage	80	200
AF1-272	PENELEC	Solar	66	110
AF1-302	PENELEC	Solar	18	42
AF1-322	AEP	Solar	84	200
AF1-330	ComEd	Solar	16.4	20
AF1-333	PPL	Solar	12	20
AF1-337	PPL	Solar	12	20
AF1-338	PPL	Solar	12	20
AF1-339	PPL	Solar	12	20
AF2-004	ATSI	Natural Gas	33	33
AF2-005	ATSI	Natural Gas	8	8
AF2-014	AEP	Solar	90	150
AF2-018	AEP	Solar	133.9	200
AF2-019	AEC	Storage	8	20
AF2-020	AEC	Storage	8	20
AF2-024	AEC	Storage	9.6	24
AF2-029	APS	Storage	8	20
AF2-030	ME	Storage	8	20
AF2-032	ComEd	Solar	13.6	20
AF2-048	AEP	Solar	46.6	70
AF2-067	Dayton	Solar	24	40
AF2-082	PPL	Solar	33	55
AF2-083	AEP	Solar	100	150
AF2-105	AEP	Solar	60	100
AF2-114	APS	Solar	9	15
AF2-116	APS	Solar	13.3	20
AF2-119	Dominion	Solar	48	80
AF2-125	AEP	Solar	35.5	54
AF2-140	PPL	Natural Gas	30	0
AF2-141	PENELEC	Storage	8	0
AF2-149	AEP	Solar	32	80
AF2-187	ATSI	Solar	78	120
AF2-191	AEP	Solar	66	110

	Transmission Owner	Fuel Type	MW Capacity (FTIR/FTWR)	MW Energy (nFTIR/nFTWR)
AF2-229	PENELEC	Solar	10.9	20
AF2-233	PPL	Solar/Storage	12	20
AF2-234	PPL	Solar/Storage	24	40
AF2-235	PENELEC	Solar	6	10
AF2-238	PENELEC	Solar	12	20
AF2-243	PENELEC	Solar	9	15
AF2-295	APS	Solar	12	20
AF2-306	EKPC	Storage	26	26
AF2-336	PENELEC	Solar	12	20
AF2-337	PENELEC	Solar	12	20
AF2-371	AEP	Solar	50.4	84
AF2-380	APS	Solar	9	15
AF2-383	ME	Solar/Storage	20	20
AF2-415	PSEG	Storage	150	150
AG1-001	AEC	Solar	11.78	31
AG1-039	ATSI	Solar	23.52	56
AG1-040	PENELEC	Solar	12	20
AG1-052	ME	Storage	20	20
AG1-054	ATSI	Solar	25.2	40
AG1-059	APS	Solar	60	100
AG1-062	JCPL	Solar	9	15
AG1-100	PENELEC	Solar	12	20
AG1-104	BGE	Storage	120	300
AG1-117	AEC	Storage	0	0
AG1-163	AEP	Solar	42	100
AG1-182	APS	Solar	65.21	100
AG1-186	APS	Storage	34.79	0
AG1-188	JCPL	Solar	8.316	19.8
AG1-189	JCPL	Solar	5.1	12
AG1-223	JCPL	Solar	6.47	15.4
AG1-254	AEC	Solar/Storage	25.3	38
AG1-281	PENELEC	Solar	12	20
AG1-301	PENELEC	Solar	12	20
AG1-307	APS	Solar/Storage	56	80
AG1-348	JCPL	Solar	8.316	19.8
AG1-415	APS	Solar	25	50
AG1-416	APS	Solar/Storage	91	125
AG1-419	ME	Solar	21	50

	Transmission Owner	Fuel Type	MW Capacity (FTIR/FTWR)	MW Energy (nFTIR/nFTWR)
AG1-420	APS	Solar	42	70
AG1-480	ME	Solar/Storage	18.2	18.2
AG1-487	JCPL	Solar/Storage	20	20
AG1-489	ATSI	Solar/Storage	145	175
AG1-495	JCPL	Solar	8.316	19.8
AG1-500	ATSI	Natural Gas	385.5	406.4
AG1-501	ATSI	Natural Gas	10	11.3
AG1-510	JCPL	Solar/Storage	12	20
AG1-511	JCPL	Solar/Storage	30	50
AG1-533	APS	Solar	24	40

Attachment D – Interconnection Network Upgrades

	Description	Cost Estimate (\$M)	Required In-Service Date
n3069	Install metering at the Keystone 345 kV substation for V3-015 project.	0.07	8/1/2012
n4106.3	Jefferson-Clifty 345 kV line sag study remediation is one location of grading to remediate clearance location of concern in span 1 to 2. Latest facility study: Extend one tower on the Jefferson-Clifty Creek (IKEC) 345 kV circuit.	0.41	12/31/2022
n4173	Un-six wire the existing Lakeview-Ottawa and Lakeview-Greenfield 138 kV lines, creating a new third line in the process, Ottawa-Greenfield 138 kV.	3.52	6/1/2015
n4287	Install 34.5 kV tap, radio controlled switch and associated equipment at North Meshoppen 34.5 kV and Tunkahannock 34.5 kV substations.	0.1	12/31/2015
n4296	The SCULL No. 2 to MILL No. 2 138 kV line: In order to mitigate the overloads, the relay settings should be adjusted at Mill 138 kV bus. The new emergency rating will be 270 MVA.	0.005	6/1/2015
n4323	Loop the Dumont-Sorenson 345 kV line into the new Eldeberry substation pending the results of the AEP sag study.	5	10/1/2014
n4601	Replace one overdutied 138 kV breaker with a 63 kA breaker at NELSON B4 bus. Breaker name: 155 15507.	2.35	6/1/2016
n4655	Reconfigure the Albright 138 kV substation to a breaker and a half configuration.	20.7	9/25/2017
n4790	Replace Dumont 345 kV substation 2500A wave trap.	0.2	6/1/2019
n4851	Glen Falls 138 kV substation: Grade and extend fence and ground grid approx. 50' x 180'. Install attachment line from 138 kV SF6 breaker to POI.	1.79	6/1/2020
n4855	Rebuild 2.5 miles of the Glen Falls to Oak Mound 138 kV line.	3.03	6/1/2020
n4997	Replace the disconnect switch at the Sabinsville 115 kV substation.	0.09	6/1/2017
n5034	Build a new Sullivan-Reynolds 765 kV line.	441.7	6/1/2021
n5119	Build a new 138 kV substation with a three-position ring bus (with provisions to add a fourth position for PJM Queue Project AB2-032) off the right-of-way for circuit 13723 in Queen Anne's County, Maryland. Two of the positions on the	7.6	11/1/2017

	ring bus will be transmission line terminals for the tie-in of circuit 13723 to the substation. The third position will be a terminal configured for the interconnection of a generator.		
n5145	Reconfigure Wilton 765 kV bus thereby allowing for 765 kV L11216 (currently on Bus 6) to be relocated to Bus 8. Along with this line relocation, installation of 2-765 kV BT CB's (6-8 and 8-2).	55.23	6/30/2019
n5146	Gilbert 230 kV substation – Construct the new 230 kV line terminal at the new Gilbert 230 kV substation for the new AB1-154 interconnection.	0.81	6/1/2020
n5165	Reconductor 0.4 miles of Gilbert-Springfield 230 kV circuit replacing 1590 ACSR with 1590 ACSS.	0.81	6/1/2020
n5171.1	Upgrade, all existing 138 kV circuit breakers at Kewanee are three-cycle devices with achievable clearing time of 6/11 cycles. Relay upgrade on line L6101 will be required to support this clearing time.	0.14	12/31/2018
n5171.2	Upgrade all existing 138 kV circuit breakers at Kewanee are three-cycle devices with achievable clearing time of 6/11 cycles. Relay upgrade on line L7411 will be required to support this clearing time.	0.14	12/31/2018
n5171.3	Upgrade all existing 138 kV circuit breakers at Kewanee are three-cycle devices with achievable clearing time of 6/11 cycles. Relay upgrade on line L7421 will be required to support this clearing time.	0.14	12/31/2018
n5171.4	Upgrade all existing 138 kV circuit breakers at Kewanee are three-cycle devices with achievable clearing time of 6/11 cycles. Relay upgrade on line L7423 will be required to support this clearing time.	0.14	12/31/2018
n5171.5	Upgrade all existing 138 kV circuit breakers at Kewanee are three-cycle devices with achievable clearing time of 6/11 cycles. Relay upgrade on line L94301 will be required to support this clearing time.	0.14	12/31/2018
n5171.6	Upgrade all existing 138 kV circuit breakers at Kewanee are three-cycle devices with achievable clearing time of 6/11 cycles. Relay upgrade on line segment of L15508 from Kewanee to U4-027 (new line number 7408).	0.14	12/31/2018
n5173	Add high-speed backup relaying and associated communications to 138 kV line Kewanee to U4-027 (7408).	2.8	12/31/2018
n5253	Reconductor the ComEd portion of Crete-St John 345 kV line.	14.9	6/1/2020
n5254	Reconductor the Lee County-Byron 345 kV line.	17.2	10/1/2017
n5254.1	Mitigate Sag on the Lee County-Byron 345 kV line.	10	9/1/2024
n5264	TOSCO_2-VFT 2 230 kV line – rebuild with paired 1033 ACSS	5.01	6/1/2021
n5357	Upgrade line protection and controls at the Vassell 765 kV substation to coordinate with the new 765 kV switching station.	1.1	3/1/2022
n5472	Sag study results on the Wildcat-Hillsboro 138 kV line show that a distribution circuit underneath the line needs to be relocated.	0.19	10/1/2020
n5520	Protection system changes at Crystal 69 kV substation.	0.12	7/1/2019
n5521	Protection system changes at Hutchings 69 kV substation.	0.5	7/1/2019
n5609	Rebuild 41.13 miles of 500 kV line 576 from Midlothian to North Anna with 3-1351.5 125C ACSR with the following ratings: Rate A: 4816, Rate B: 4816 and Rate C: 5539.	234.441	12/1/2021

n5612	Rebuild 2.8 miles 768 ACSS with a rating of A: 393, B: 393, C: 452 at AC1-076 Tap-Paytes DP 115 kV line.	3.65	6/30/2019
n5613	Rebuild or reconductor 0.9 miles of overhead conductor (ACSR ~ 397.5 ~ 30/7 ~ LARK) between ALTUSTA and 05OTTER 138 kV line.	2.74	10/1/2019
n5613.1	Replace 795 AAC station conductors at Altavista 138 kV substation.	0.1	1/21/2028
n5668	Install new 138 kV three-breaker ring bus station along the Rio-Lick 138 kV line, installation of associated protection and control equipment, line risers, switches, jumpers and supervisory control and data acquisition (SCADA) equipment.	6.7685	10/31/2016
n5669	Modify Rio to Lick 138 kV transmission line cut in.	1.0727	10/31/2016
n5670	Install 138 kV revenue meter, generator lead transmission line span from the new Terrapin 138 kV station to the point of interconnection, and install two diverse fiber-optic paths from the point of interconnection to the Terrapin 138 kV station control house.	0.8305	10/31/2016
n5674	Rebuild and expand the existing Elk 138 kV station by removing the two existing 138 kV box bays and associated equipment and installing a new five breaker 138 kV station, physically designed as a six-position breaker and a half station but operated as a five-breaker ring bus. The two existing 138 kV circuit breakers will be reused and three new 138 kV circuit breakers will be installed. A 16' x 12' DICM expansion will be added to existing Elk 138 kV station control house. Install other associated line protection and control equipment, 138 kV line risers, switches, jumpers and supervisory control and data acquisition (SCADA) equipment. Review and update remote end protection relay settings at the Corwin and Lemaster 138 kV stations.	6.722	10/31/2016
n5675	Reroute and reterminate the Corwin-Elk and Elk-Lemaster 138 kV circuits to a new bay position in the rebuilt Elk 138 kV station.	0.6951	10/31/2016
n5676	Install portion of 138 kV generator lead line from the Elk 138 kV station to the point of change of ownership with interconnection customer, 138 kV revenue meter, and dual fiber-optic cable from the Elk 138 kV station control house to interconnection customer's fiber connection/transition.	0.9501	10/31/2016
n5699	Install 138 kV revenue meter, generator lead transmission line span from the new Red Run 138 kV station to the point of interconnection, and install two diverse fiber-optic paths from the point of interconnection to the Red Run 138 kV station control house.	0.612	3/21/2019
n5700	Install new 138 kV three-breaker ring bus station along the Chatfield-Howard 138 kV line, installation of associated protection and control equipment, line risers, switches, jumpers and supervisory control and data acquisition (SCADA) equipment.	5.4417	3/21/2019
n5701	Chatfield-Crawford 138 kV T-line cut in	1.1041	3/21/2019
n5702	Install new SFP transceivers and update protective relay settings at Chatfield and Howard 138 kV stations.	0.0794	3/21/2019
n5703	Install one fiber-optic path to the AEP Network to facilitate relaying between Red Run, Howard and Melmore 138 kV stations.	0.5592	3/21/2019
n5780	Reconductor Stuart-Spurlock 345 kV line with twin bundle 1033 Curlew ACCR conductor.	17	7/1/2022

n5788	Substation reinforcements at Preston substation and at Todd 69 kV substations. The estimate to perform this work is \$67,000 and will take approximately one year to complete. The new emergency rating would be 173 MVA.	0.1	12/14/2018
n5858	Increase MOT on Headquarters-Snow Hill 69 kV line.	0.25	6/1/2020
n5859	Increase MOT on Snow Hill-Murphysville 69 kV line.	1.1	6/1/2020
n5917	Reconductor the E Frankfort-Crete 345 kV line.	10.3	6/1/2022
n6037	Expansion of Sullivan 345 kV substation and installation of associated protection and control equipment	1.75	6/1/2019
n6037.1	Install breakers, physical structures, protection and control equipment, and associated facilities at Sullivan 345 kV switching station.	3.5746	6/1/2019
n6037.2	Install custom dead-end transmission line structure to raise Sullivan-West Casey 345 kV line.	0.6119	6/1/2019
n6037.3	Install custom dead-end transmission line structure to raise Sullivan-Petersburg 345 kV line.	0.6119	6/1/2019
n6038	Install 345 kV revenue metering at Sullivan 345 kV switching station.	0.35	6/1/2019
n6039	Expansion of Sullivan 345 kV substation and installation of associated protection and control equipment	1.75	6/1/2019
n6039.1	Install breakers, physical structures, protection and control equipment, and associated facilities at Sullivan 345 kV switching station.	1.6773	6/1/2019
n6040	Install 345 kV revenue metering at Sullivan 345 kV switching station	0.35	6/1/2019
n6092	Install attachment facility line and associated hardware to accept the interconnection customer generator lead line terminating at the AD2-171 interconnection substation and install revenue metering.	2.062	6/1/2023
n6093	Install a new 500 kV three-breaker ring bus substation along the Alburtis-Branchburg 500 kV line.	31.2	6/1/2023
n6094	Loop the Branchburg-Alburtis 500 kV circuit into the AD2-171 interconnection switching station.	36.286	6/1/2023
n6095	Modify the existing Alburtis 500 kV circuit breaker protection and control scheme and tie breaker protection and control scheme.	0.155	6/1/2023
n6096	Modify the existing Branchburg 500 kV circuit breaker protection and control scheme and Branchburg-Breinigsville 500 kV tie breaker protection and control scheme.	0.155	6/1/2023
n6115	For DEV portion, rebuild 4.7 miles of 115 kV line 45 from Kerr Dam to GW King tap with 768 ACSS.	11.3505	10/31/2020
n6124	Increasing the Danville-East Danville 138 kV circuit summer rating to 337/482 MVA will still require AEP to rebuild the line.	8.25	6/1/2021
n6130	Rebuild 12.4 miles 2-636 ACSR between Chesterfield-Basin 230 kV, ratings after the upgrade: 1047/1047/1204.	31	12/1/2021
n6137	Perform a sag study and mitigation on the AEP-owned portion of the Tidd-Wylie Ridge 345 kV tie-line circuit (8.6 miles). This will raise the operating temperature and ratings to the standard value.	0.035	2/1/2022
n6141	Replace Relay at Everetts 115 kV, rating after the upgrade: 300/300/345.	0.5	6/1/2021

n6142	Rebuild 5.14 miles 636 ACSR between AB2-169 TAP 115.0 kV and 3FIVE PT 115.0 kV circuit 1. Rating after the upgrade: 261/261/301.	6.682	6/1/2021
n6144	Rebuild DEV portion 20.5 miles 2-636 ACSR and rebuild 17.5 miles 2-636 ACSR between 6GREENVILE T-6EVERETS 230.0 kV substations. Rating after the upgrade: 1047/1047/1204.	39.92	6/1/2019
n6178	Rebuild 4.5 miles of 230 kV line 2003 from Harrowgate to Tyler with 2-636 ACSR.	11.25	12/31/2024
n6179	Rebuild 12.3 miles of 230 kV line 2003 from Poe to Harrowgate with 2-636 ACSR.	30.75	12/31/2024
n6185	Build a new 138 kV line from Black River to Astor substation.	18.67	6/1/2022
n6238	Increase the operating temperature of the 556.5 MCM ACSR/TW conductor from 212F to 302F at Central Hardin 69 kV substation.	0.04	12/31/2024
n6238.4	Replace the 750 MCM copper jumpers at the Central Hardin 69 kV substation using bundled 500 MCM copper or equivalent.	0.01	11/30/2026
n6238.5	Replace the 1200A disconnect switches W124-623 and W124-625 at Central Hardin 69 kV substation and W80-605 at the Kargle tap location.	0.31	11/30/2026
n6263.2	Replace 600 Amp switch at Harrison to upgrade the Harrison-Zuber 138 kV line.	0.2	12/31/2020
n6287	Trowbridge 230/115 kV substation reconfiguration. 230 kV yard changes: Expand 230 kV yard to accommodate additional rung of 230 kV breakers. This will require relocation of existing 115 kV equipment. Relocate transformer No. 1 to accommodate new rung of 230 kV breakers. Install one new 230 kV breaker, and two new 230 kV backbones. Rereminate line 2126 on new rung – will require T-line reconfiguration outside of station. Terminate new AD1-074/75/76 line on new rung. Will require T-line reconfiguration outside of the station to utilize existing 115 kV line 25 station entrance infrastructure. 115 kV yard changes: Expand station to the north and rebuild entire 115 kV yard (eight breakers, four backbones). Rereminate all four existing 115 kV lines line 1020, and line 25 will require significant T-line reconfigurations outside of the station.	8.1	6/1/2021
n6287.1	Add additional 230 kV breaker at Trowbridge to prevent loss of TX No. 1 upon fault on 230 kV line 2034 in stuck breaker scenario.	0.62	6/21/2021
n6288	A preliminary estimate for sag mitigation between GREENACRE; T and 05OLIVE 345 kV is \$13.9M with an estimated construction timeline of 30 months.	13.9	9/1/2021
n6328	Reconductor 0.64 miles of 230 kV line 205 from Harrowgate to Tyler with 1033.5 ACSR. Uprate 4.09 miles of 230 kV line 205 from Harrowgate to Tyler.	2.148	6/1/2022
n6379	Rebuild 6.47 miles of 230 kV line 2028 from Mt Eagle to Charlottesville with 2-636 ACSR.	16.175	10/1/2026
n6401.1	Rebuild 13 miles of Adkins-Beatty 345 kV line.	55	12/1/2022
n6461	Perform sag study on AEP's portion of Wildcat-Kenton (Kenton-Emerss) 138 kV circuit, 23.5 miles 477 ACSR 26/7 Hawk conductor.	0.094	6/1/2022
n6463.1	Increase MOT of Boone-Longbranch 138 kV line section 954 MCM conductor to 275F (~2.25 miles).	0.2	6/1/2022
n6463.2	Upgrade bus and jumpers associated with Boone 138 kV bus using 2-500 MCM 37 CU conductor or equivalent on the Boone Co-Longbranch 138 kV line.	0.17	6/1/2022
n6480	Increase the maximum operating temperature of the 266 MCM ACSR conductor in the Murphysville-Plumville 69 kV line section to 266 degrees F (9.9 miles).	0.65	6/1/2022

n6497.2	Replace 6, 3000A Rockport CTs on the Rockport-Jefferson 765 kV line.	4.8	12/1/2022
n6514	Line No. 2068 Halifax (Sedge Hill)–AD1-087 tAP 230 kV: wreck and rebuild the line of 16 miles to increase its line rating to 1047 MVA (normal), 1047 MVA (emergency), and 1204 MVA (load dump). A SCC and a Va CPCN is required.	25.07	12/31/2021
n6539	Build new 500 kV line from Rawlings to Morrisville substation 110 miles.	627	12/31/2021
n6574	Rebuild 14.53 miles of 500 kV line 575 from North Anna to Ladysmith with 3-1351.5 113C ACSR.	45.043	12/31/2021
n6580	Rebuild 16.45 miles of 500 kV line 585 from AE2-094 Tap to Carson with 3-1351.5 125C ACSR.	50.995	6/1/2022
n6587	Reconductor 0.08 miles of Oyster Creek-Cedar 230 kV line transmission line. Upgrade Cedar Terminal of Oyster Creek 230 kV substation.	0.64519	6/1/2022
n6593	Replace Adam substation conductor on the AE2-089 Tap-Adam 138 kV line.	0.3	12/31/2022
n6595	Relay Modifications at the Harwood 69 kV substation.	0.16	12/31/2020
n6634.1	New 138 kV switching station – Construct a new three circuit breaker 138 kV switching station physically configured in a breaker and half bus arrangement but operated as a ring-bus for AC2-061 interconnection.	6	7/1/2020
n6634.2	Install 138 kV revenue metering at Hillsboro-Clinton County 138 kV.	0.25	7/1/2020
n6634.3	Install Hillsboro-Clinton County 138 kV transmission line cut in.	1	7/1/2020
n6634.4	Upgrade line protection and controls at the Hillsboro 138 kV substation to coordinate with the new AC2-061 138 kV switching station.	0.25	7/1/2020
n6634.5	Re-tune wave traps at two (Hillsboro and Clinton County 138 kV) substations and adjust relay settings at one.	0	7/1/2020
n6643	Reconductor 0.3 miles of the Old Church 34.5 kV circuit.	0.14	12/11/2019
n6676.1	EKPC to install necessary equipment (a 69 kV isolation switch structure and associated switch, plus interconnection metering, fiber-optic connection and telecommunications equipment, circuit breaker and associated switches, and relay panel) at the new East Harrison 69 kV switching station to accept the IC generator lead line/bus.	0.73	6/1/2022
n6676.2	EKPC to construct a new 69 kV switching station (East Harrison switching) to facilitate connection of the Blue Moon Solar generation project (AD2-048).	2.77	6/1/2022
n6676.3	EKPC to construct facilities (~250 feet) to loop the existing Cynthiana Tie-Headquarters 69 kV line section into the new East Harrison switching substation.	0.13	6/1/2022
n6676.4	EKPC to modify relay settings at Renaker 69 kV substation for existing line to East Harrison 69 kV switching station.	0.01	6/1/2022
n6676.5	EKPC to upgrade relays and modify relay settings at headquarters 69 kV substation for existing line to East Harrison 69 kV switching substation.	0.01	6/1/2022
n6676.6	EKPC to install OPGW in the Renaker-3M-Cynthiana Tie 69 kV line sections (9.3 miles).	0.45	6/1/2022
n6680	Construct one span 25 kV tap to customer's substation on Young Jct. to Sligo 25 kV line.	0.03	6/1/2023
n6681	Install 25 kV metering in the customer's substation at Young Jct. to Sligo 25 kV line.	0.0389	6/1/2023

n6682	Construct 25 kV line tap with two 25 kV switches on the at Young Jct. to Sligo 25 kV line for the interconnection of AF1-210.	0.0625	6/1/2023
n6683	Update relay settings at Burma 23 kV.	0.0144	6/1/2023
n6684	Update relay settings at Templeton 23 kV.	0.0144	6/1/2023
n6712	Install AC2-195 ADSS fiber: from the new AC2-195, interconnection substation to the anticipated ADSS cable near the intersection of Marion Williamsport Road and N Main Street proposed for PJM queue position AB2-131. The assumed route is a combination of aerial ADSS (0.87 miles) and underground bore (0.14 miles).	0.165	12/31/2020
n6759.2	Replace two 1600 A switches at Dearborn and four switches at Pierce on the Dearborn-Pierce 345 kV line.	9	12/31/2022
n6769.1	Upgrade/replace three 345 kV 1600A switches at Beatty station.	1.5	8/31/2026
n6840	ComEd 345 kV L11212 SSTE rating is 1846 MVA. The upgrade will be to install a new 345 kV bus tie circuit breaker at station 12 Dresden. The new 345 kV breaker will be installed as BT CB 12-13. Initial review of this proposal is that the existing contingency will be reduced and potentially reducing the post-contingency flow. PJM to confirm this proposal in study. The ratings for L11212 will not change rather the contingency as stated above will be revised.	3.44	11/24/2026
n6865	Construct a tap and install three 34.5 kV load-break air switches on the West Flemington Tap-Frenchtown Solar 2 portion of the East Flemington-Frenchtown (A729) 34.5 kV line.	0.403	4/17/2020
n6866	Install FE owned 34.5 kV metering in the interconnection customer's substation for AF2-192.	0.27	4/17/2020
n6867	Upgrade relaying at East Flemington 34.5 kV.	0.403	4/17/2020
n6868	Upgrade relaying at Frenchtown 34.5 kV.	0.403	4/17/2020
n6919	Tap the existing Cambridge Springs 34.5 kV line at an existing pole or interspersed pole on Penelec's existing distribution circuit (00245-52) near pole X-67952, new SCADA recloser tap to interconnect queue project AF1-217. Install 34.5 kV metering in customer's facilities. The customer will have to provide Penelec with permanent access/roadway to this off-road location/equipment. The customer is responsible to build their own line from their site to Penelec's existing facilities.	0.1311	3/1/2021
n6920	Edinboro South 34.5 kV SS. Adjust remote relay and metering settings.	0.0107	3/1/2021
n6921	Review customer drawings, create nameplates and update CD drawing at Edinboro 34.5 kV.	0.0223	3/1/2021
n6954	Install a 138 kV steel isolation switch structure and associated 138 kV isolation switch, conductor from the terminal structure to the point of interconnection, interconnection metering, fiber-optic connection, line differential relay panel and telecommunications equipment to accept the IC generator lead line at Avon 138 kV substation.	0.6978	12/1/2022
n6955	Modify Avon 138 kV substation by installing two circuit breakers, A-frame structure and associated terminal switches, bus conductor, PTs, foundations, breaker relay panel for new bus position; a new steel structure (for repositioned Dale line); transfer existing conductor, OPGW fiber optic and existing relay panel on Avon-Dale line at Avon substation.	1.7632	12/1/2022

n7051	To mitigate the Lewis 138 kV bus tie overload, breaker “JN” will be replaced with a 3000-amp breaker.	0.236	5/31/2023
n7127.2	Reconductor 13.01 miles of line East Towanda to Canyon 230 kV line. Replace transmission line drop at East Towanda.	89.22415	5/31/2022
n7127.3	Replace disconnect switch at Canyon 230 kV substation.	1.05697	5/31/2022
n7127.4	Replace line trap at East Towanda 230 kV substation.	0.225	5/31/2022
n7127.5	Adjust/replace metering and line relaying at East Towanda 230 kV substation.	0.5	5/31/2022
n7137	Reconductor METUCHEN-NEWDORV_H 230 kV transmission line to 1080 MVA SER.	12.88	6/1/2022
n7180	Rebuild 7.2 miles of 230 kV line 235 from Prince EDW to Farmville with two 768.2 ACSS/TW (20/7) “MAUMEE” at 250 C.	20.7	12/31/2022
n7202	Reconductor 0.1 miles of the Chapman to AA2-000 tap 230 kV line.	0.45	5/30/2022
n7290	Construct generator lead first span exiting the POI station, including the first structure outside the fence at Lockwood Road 138 kV.	0.4	9/28/2017
n7291	Install 138 kV revenue metering at Lockwood Road 138 kV substation.	0.388	9/28/2017
n7292	Upgrade line protections and controls at the Lockwood Road 138 kV substation.	0.045	9/28/2017
n7293	Expand the Southwest Lima 345 kV substation: Install one additional 345 kV circuit breaker and extend the 345 kV bus. Installation of associated protection and control equipment, 345 kV line risers and SCADA.	1.762	9/20/2019
n7294	Construct generator lead first span exiting the point of interconnection station, including the first structure outside the fence at South West Lima 345 kV.	1.025	9/20/2019
n7295	Install 345 kV revenue metering at South West Lima 345 kV substation.	0.427	9/20/2019
n7310	AE2-285 generator lead termination: Installation of foundations, disconnect switch and associated equipment to accommodate the termination of the 69 kV generator lead line at the new AE2-285 substation.	0.0917	6/30/2022
n7311	Metering: Install customer-owned revenue metering at the new AE2-285 substation.	0.0025	6/30/2022
n7383	Relaying upgrades at TSS 100 Shady Oaks 138 kV substation including: Install a SEL-411L as current differential line protection on L94701 and make the existing primary relay, SEL-311L, the secondary relay. Modify L16901 CB and L13901 CB tripping to accommodate new topology. Install load rejection logic such that transfer trip is initiated on both primary and secondary relaying to TSS 946 GSG-6 Wind Farm if L94701 CBs are open at TSS 100 Shady Oaks.	0.4101	12/31/2020
n7448	Upgrade existing Varner 138 kV station.	1.772288	4/3/2023
n7554.1	Replace five sub conductor 2000 AAC 91 Str at Danville2 138 kV station.	0	12/31/2024
n7554.2	Replace three sub conductor 2000 AAC 91 Str at East Danville 138 kV station.	0	12/31/2024
n7599	Reconductor one section of 4/0 CU sub conductor circular at Drakes town 115 kV with a conductor able to meet or exceed 254 MVA STE. Reconductor 14.18 miles of 556.5 ACSR 26/7 115 kV transmission line from Belvedere Tap to Drakes town with 795 ACSR or a conductor able to meet or exceed 254 MVA STE. Replace one 5 A thermal relay at Pequest River.	49.6295	12/17/2021
n7600	Replace one 800 A Generic wave trap at Whippany 115 kV.	0.1343	12/17/2021

n7754.3	The Danville-East Danville 138 kV circuit will require higher capacity conductor to obtain a rating higher than the 482 MVA achieved by the initial rebuild. As such, APCo will instead install 1272 ACSS conductor during the initial rebuild project under n6124.	0.46	6/1/2024
n7854.1	Rearrange line No. 65 to loop into and out of the new three breaker AE1-155 115 kV switching station. Line 65 is an existing 115 kV line that runs from Northern Neck substation to Harmony Village substation. AE1-155 provides for the construction of a new substation located in the existing line 65 right-of-way between existing structures 65/498 and 65/499 in Farnham, Virginia.	1.0672	9/15/2022
n7854.2	Build a three breaker AE1-155 115 kV switching station. The facilities identified provide for the initial construction of a new 115 kV three-breaker ring substation between structures 65/498 and 65/499.	5.4139	9/15/2022
n7854.3	Remote protection and communication work. Additional work is required at Northern Neck, Rappahannock and Harmony Village 115 kV substations. Drawing work, relay resets and field support necessary to change the line 65 destinations at Garner DP, Lancaster, Ocran and White Stone substations will also be completed.	0.3182	9/15/2022
n7881	Perform a sag study on the Dearborn-Pierce 345 kV line transmission line. OVECs cost estimate for performing the sag study is \$125,000. Sag study results (from AE2-297 facilities study) show the need to replace 16 tangent structures, three dead-end structures, and conductor over one Ohio river crossing. Cost estimate is \$11.383 M. New SE rating to be 1165 MVA.	11.38	12/31/2021
n7923	Harwood 69 kV: Model IC in CAPE and conduct a wide area short circuit study two busses away from the IC facilities. Identify affected relays and revise settings as needed. Conduct a detailed review of the IC relay settings and engineering packaged submitted by IC to the PPL EU. Modify relays accordingly.	0.123	12/31/2020
n7926	Tap the Maysville-Sharon 69 kV (Y-301) line near structure 62 and install disconnect switches on either side of the tap at the new AE1-079 substation. Includes project management, environmental, forestry, real estate and right-of-way.	1.193	6/1/2020
n7927	Update relay settings at Marysville 69 kV substation.	0.0926	6/1/2020
n7928	Update relay settings at Sharon 69 kV substation.	0.0926	6/1/2020
n7929	Update relay settings at Masury 69 kV substation.	0.0926	6/1/2020
n7930	Update relay settings at Cedar street 69 kV substation.	0.0926	6/1/2020
n7931	Update relay settings at McDowell Street 69 kV substation.	0.0926	6/1/2020
n7932	Update relay settings at Dilworth 69 kV substation.	0.0926	6/1/2020
n7933	SCADA/fiber communication: Estimated installation of 700 MHz radio system (70% penetration of FE territory) to support the SCADA switch replacements at the new AE1-079 substation.	0.1033	6/1/2020
n7934	Tap the Dilworth-Andover (Maysville) 69 kV (Y-78) line at structure 145 and install disconnect switches on either side of the tap. Includes project management, environmental, forestry, real estate and right-of-way.	0.883	6/1/2020
n7935	Update relay settings at Marysville 69 kV substation.	0.0926	6/1/2020
n7936	Update relay settings at Sharon 69 kV substation.	0.0926	6/1/2020

n7937	SCADA/fiber communication: Estimated installation of 700 MHz radio system (70% penetration of FE territory) to support the SCADA switch replacements at AE1-237 substation. Assumed SCADA work is included in this cost.	0.0517	6/1/2020
n7951	Greenfield 69 kV substation – Convert the existing 69 kV substation to a two-bus configuration substation. This includes the installation of five 69 kV circuit breakers, disconnect switches, all physical structures, protection and control equipment, communications equipment, and associated facilities at the Greenfield 69 kV substation.	1.315	12/31/2021
n7995	AE2-117: Tap the Warrior Ridge-Williamsburg 46 kV line to provide interconnection facilities to the customer substation, approximately 2.5 miles from the Alexandria substation. Install SCADA-controlled MOAB switches on either side of tap location.	1.0047	10/2/2021
n7996	Williamsburg 46 kV: Replace existing line relaying and control panel for Tyrone North-Warrior Ridge line exit with one standard prewired relaying panel.	0.2508	10/2/2021
n7997	Warrior Ridge 46 kV: Replace existing line relaying and control panel for Tyrone North-Williamsburg line exit with one standard prewired relaying panel.	0.2508	10/2/2021
n7998	Tyrone North 46 kV: Replace existing line relaying and control panel for Warrior Ridge-Williamsburg line exit with one standard prewired relaying panel.	0.2508	10/2/2021
n8007	Upgrade line protection and controls at the Cherry Creek 138 kV station.	0.0464	9/25/2018
n8008	Install fiber from the new 138 kV station to the Grandview 138 kV station for relaying.	0.1463	9/25/2018
n8026	Install new underground fiber between IC switching station and ring bus substations control enclosure at AE1-179 South Millville-Newport 69 kV line.	0.2	11/1/2021
n8032.1	Tap the Larrabee-Point Pleasant No 1 34.5 kV line approximately 1.9 miles from Larrabee and 3.9 miles from Allaire to the customer facility and build a single span to point of interconnection with customer.	2.2647	2/18/2022
n8032.2	Review relay settings at Point Pleasant 34.5 kV.	0.0439	2/18/2022
n8032.3	Review relay settings at Larrabee 34.5 kV.	0.0351	2/18/2022
n8070.3	This project provides for the initial construction of a new 230 kV three-breaker ring substation between transmission structures 2034/220 and 2034/221. The objective of this project is to build a 230 kV, three-breaker ring bus to support the new 120 MW solar farm built by Sumac Solar, LLC. The site is located along Dominion Energy's existing 230 kV, 2034 line from Earleys to Trowbridge. The cut line will consume two of the positions in the ring bus. The third position will be for the 230 kV feed from Sumac Solar, LLC collector station for the new 120 MW solar farm.	6.4317	6/1/2019
n8071.1	Construct a new 138 kV switching station for AF2-083 interconnection.	2.0866	10/21/2022
n8071.2	Install transmission line cut-in and update protection settings at Kenzie Creek and Stone Lake 69 kV substations.	0.342	10/21/2022
n8071.3	Install fiber-optic connections at Kenzie Creek and Stone Lake 69 kV substations.	0.0495	10/21/2022
n8071.4	Replace substation conductors at Stoner Lake 69 kV.	0.0391	10/21/2022
n8071.5	Replace 16 structures and associated spans of conductor at Kenzie Creek and Stone Lake 69 kV substations.	0.8699	10/21/2022

n8073	Install relaying at Kewanee 138 kV substation for the new bay position. Conduct a detailed review of the IC relay settings.	0.603	12/31/2019
n8076.1	Install a 69 kV switch structure, associated switch and 69 kV breaker, disconnect switches, interconnect metering, relaying and fiber at new AF1-038 switching station to accept the IC's generator line.	1.55	6/6/2019
n8076.2	AF1-038 69 kV interconnect switching station (North Russell County Switching) – Construct a new 69 kV switching station to connect the Queue No. AF1-038 solar generating facility.	6.11	6/6/2019
n8076.3	Sewellton Junction-Windsor 69 kV – Construct a new loop-in tap line (~500 ft.) from EKPC's existing Sewellton Junction-Windsor 69 kV line to the new AF1-038 69 kV interconnection switching station.	0.99	6/6/2019
n8076.4	Sewellton Junction 69 kV substation – Upgrade relays and modify relay settings.	0.345	6/6/2019
n8076.5	Windsor 69 kV substation – Upgrade relays and modify relay settings.	0.345	6/6/2019
n8076.6	Install OPGW fiber on the 69 kV line from Sewellton Junction to the new North Russell County Switching station, approximately one mile in length.	0.57	6/6/2019
n8077.1	New switching station (South Green County switching station) to interconnect customer facility	4.095	12/31/2022
n8077.2	Loop-in tap line to new AF1-050 switching station from Summer Shade-Green County 161 kV line.	0.276	12/31/2022
n8077.3	Replace existing and modify relay settings at Summer Shade 161 kV substation.	0.01	12/31/2022
n8077.4	Upgrade relays and modify relay settings at Green County 161 kV substation.	0.01	12/31/2022
n8077.5	Install OPGW between Green County-South Green County 161 kV switching stations.	0.726	12/31/2022
n8087.1	McDowell 69 kV: Replace Maysville Y-80/107 line relay and control panel with a standard line relaying panel.	0.7186	8/15/2024
n8087.2	Masury 69 kV: Relay settings will be revised.	0.0404	8/15/2024
n8087.3	AF1-093: Tap the Maysville-McDowell 69 kV line roughly 8.2 miles from McDowell substation, near structure 108. Install two 69 kV line switches with SCADA. Install one span of 795 ACSR Drake toward the customer substation.	0.7721	8/15/2024
n8087.4	Maysville 69 kV: Relay settings will be revised.	0.0388	8/15/2024
n8087.5	Cedar Street 69 kV: Relay settings will be revised.	0.0388	8/15/2024
n8087.6	Sharon 69 kV: Relay settings will be revised.	0.0388	8/15/2024
n8087.7	Campbell 69 kV: Relay settings will be revised.	0.0388	8/15/2024
n8089.1	Expand Huntington Jct. switching station into a four-breaker ring bus.	4.2344	6/1/2022
n8089.2	Sorenson-Delaware 138 kV reroute, Sorenson and Hummel Creek 138 kV line reterminations	0.3849	6/1/2022
n8089.3	Install transition fiber cable at Huntington Jct. 138 kV substation.	0.1526	6/1/2022
n8089.4	Sorenson and Hummel Creek 138 kV remote end work.	0.0471	6/1/2022
n8158	Expand the Jug Street 138 kV station, including the addition of one 138 kV circuit breaker, installation of associated protection and control equipment, line risers, switches, jumpers and supervisory control and data acquisition (SCADA) equipment.	0.9386	7/31/2019

n8162	ComEd 345 kV L0304 SSTE rating is 1512 MVA. The upgrade will be to mitigate the sag on the Powerton-Tazewell 345 kV line and replace a 345 kV disconnect switch. A preliminary estimate for this work is \$9.7M with an estimated construction timeline of 30 months. Upon completion of this upgrade, the new ratings will be 1679/2058/2107/2280/2622 MVA (SN/SLTE/SSTE/SLD/ALDR).	9.7	1/18/2027
n8167.1	Rearrange line No. 2056 to loop into and out of the new three-breaker AD1-056/AD1-057 230 kV switching station.	1.7083	8/14/2017
n8167.2	Build a three-breaker AD1-056/AD1-057 230 kV switching station.	7.1907	8/14/2017
n8167.3	Remote protection and communication work at Hathaway 230 kV and Hornertown 230 kV substations	0.0666	8/14/2017
n8177	Install one dual SEL-311L relay protection package, fiber connected to the customer end, for primary and backup protection of the interconnect line at the Woodsdale 345 kV substation.	0.0371	3/28/2019
n8185	Rebuild approximately 3.5 miles of the 0762 South Millville to Newport 69 kV line from South Millville to the point of interconnection substation utilizing steel monopoles, 795 ACSR conductor and OPGW.	9.9	11/1/2021
n8189.7	Replace and/or modify remote relay communication devices and SCADA equipment at Powerton-Nevada 345 kV line to reliably interconnect the customer facility with the transmission system at the applicable remote end stations. Remove microwave equipment at remote end stations.	0.809386	4/6/2026
n8195.9	Relay upgrades at TSS 112 Wilton Center 345 kV.	0.29	9/30/2020
n8196.1	AEP will replace two existing double circuit towers with steel double circuit single pole structures on the Allen-RPMone 345 kV line. New rating is 1154 MVA Rate B. Scope is from the facility study results of the N6740 Sag Study for AE1-113.	0.9164	1/31/2025
n8201	Increase the maximum operating temperature of the 266 MCM ACSR conductor in the 2Webb CR R T line section to 266 degrees Fahrenheit.	0.135	1/31/2025
n8208.1	Construct a new three circuit breaker 138 kV station, Atalla 138 kV station, physically designed as a breaker and half bus station but operated as a three-breaker ring bus.	5.9239	12/31/2022
n8208.2	Install two structures, two spans of conductor and OPGW shield wire, connect Atalla 138 kV station to existing transmission circuit.	0.7743	12/31/2022
n8208.3	Install two fiber-optic paths to facilitate SCADA/CES connectivity; includes CES ICON upgrade work at Adams and Jay stations 138 kV.	0.198	12/31/2022
n8283	Install one new interconnect at Chesterfield 230 kV station.	4.105	12/1/2025
n8311.1	Install line 149 tap to AE2-259 switching station.	6.0705	9/15/2021
n8311.2	Remote protection and communication work at Curdsville 115 kV substation	0.1472	9/15/2021
n8317.1	Expand Fenress 500 kV and 230 kV substations to terminate three new 230 kV lines from Harpers Road substation and connect them to Fentress 500 kV.	157.5761	12/31/2025
n8317.2	Construct three overhead 230 kV transmission lines that will start at the existing Fentress substation and terminate at the new Harpers Road substation, located approximately 14.3 miles away.	150.6418	12/31/2025
n8317.3	Build new 230 kV Harpers Road switching station to terminate nine offshore wind export cables for AF1-123/124/125 and consolidate to three overhead 230 kV transmission lines to go to Fentress substation.	274.9504	12/31/2025

n8409	For the Napoleon Muni-Midway 138 kV line, reconductor the section of 336 ACSR 30/7 of the transmission line with 556.5 ACSR. Change the relay settings of the SEL relay for the ZR component to meet or exceed 262 MVA for summer emergency.	23.1006	3/31/2022
n8420	Install one new 138 kV circuit breaker, associated equipment, update protective relay settings at the Jug Street 138 kV station.	0.9386	8/22/2022
n8432.1	Rearrange line No. 269 to loop into and out of the new three-breaker AE1-072 230 kV switching station.	3.89	6/1/2026
n8432.2	Remote protection and communication work at Fentress and Sharworo 115 kV substation	0.433	6/1/2026
n8432.3	Build a three breaker AE1-072 230 kV switching station.	1.058	6/1/2026
n8435	Update protective relay settings at the Platter Creek 69 kV station.	0.045	12/1/2021
n8436.1	Reroute existing L93913 to the new termination at the new 345 kV Bus 4 at TSS 908 Mole Creek 345 kV.	3.571	12/31/2021
n8436.2	Expand TSS 908 Mole Creek 345 kV to a six-breaker ring bus.	17.659	12/31/2021
n8446.1	Modify relays at Dauphin 69 kV substation.	0.137	12/31/2027
n8446.2	Modify relays at Sunbury 69 kV substation.	0.137	12/31/2027
n8446.4	Install a new single circuit, custom foundation, custom steel pole, full terminal dead-end structure (PCO) between Sunbury-Dauphin 69 kV line. The PCO structure must be outside of the Project Developer's substation fenced area.	0.01	12/31/2027
n8448.1	Install two structures, two spans of conductor; connect ITO 69 station to existing transmission circuit.	0.68	12/31/2021
n8448.2	Install two fiber-optic paths to facilitate relaying between ITO, Haviland and South Van Wert 69 kV stations.	0.2632	12/31/2021
n8448.3	Replace protective relays, other equipment at Haviland and South Van Wert 69 kV stations.	0.7422	12/31/2021
n8448.4	Review/update protective relay settings at the proposed AE2-298 69 kV station.	0.4533	12/31/2021
n8454	Lower the Southwest Lima-West Moulton 138 kV circuit.	0.264	9/20/2019
n8456	Review the protection and control settings at the Elk 138 kV station and adjust as needed. Review the Elk 138 kV station relay settings.	0.0471	12/31/2021
n8459.1	Install custom steel double circuit tap structure, two conductor spans for in-and-out from the new 138 kV station to existing Smith Mountain-East Danville 138 kV circuit.	1.28	9/30/2021
n8459.2	Remote protection and relay work at Museville 138 kV.	0.421	9/30/2021
n8459.3	Remote protection and relay work at East Danville 138 kV.	0.421	9/30/2021
n8459.4	Build a three-breaker at AE1-250 138 kV switching station.	4.286	9/30/2021
n8474.1	Install four structures, four spans of conductor; connect the proposed 345 kV AF1-228 station to existing Beatty-Greene 345 kV transmission circuit (AEP).	2.7775	10/31/2022
n8474.2	Update protective relay settings at the Beatty and proposed AE2-148 345 kV (AEP) substations.	0.0665	10/31/2022
n8474.3	Review and revise (as needed) the Greene 345 kV station remote end line protection relay settings depending on the relative timing of completion of AE2-148 and AF1-228 (Dayton-AES Ohio).	0.25	10/31/2022

n8475.1	Install three structures, three spans of conductor with shield wire, connect proposed 345 kV station to existing Galion to South Berwick 345 KV line.	1.877	10/31/2022
n8475.2	Replace protective relays at South Berwick 345 kV station, install SFP transceiver, remove wave trap.	0.3238	10/31/2022
n8475.3	Install one six-mile fiber-optic cable path to facilitate direct fiber relaying and SCADA connectivity between the proposed AF1-229 345 kV station and Berwick station.	0.463	10/31/2022
n8475.4	Review and revise (as needed) the Galion 345 kV station remote end line protection relay settings.	0.25	10/31/2022
n8477	Build a new 230 kV interconnect into Finneywood substation. This will add a new line position and one new 230 kV breaker installed at Finneywood substation to support the new Solar Farm built by 7 Bridges Solar LLC. Finneywood substation will be a four-bay breaker-and-a-half substation located along the 230 kV 235 line between Briery and Clover and one of the line positions will be for 7 Bridges Solar.	1.1781	12/31/2025
n8478	This project serves to install one new 230 kV backbone structure, one new 230 kV self-supporting switch, and one 230 kV three-pole structure at the new Finneywood 230 kV substation. The backbone shall be installed inside the substation, and the three-pole structure shall be installed outside of the substation allowing 7 Bridges Solar to connect into Finneywood substation. The conductor and shield wire to be used will be a bundled conductor 768.2 ACSS/TW/HS "Maumee" and dual DNO-1141- OPGW respectively.	2.1547	12/31/2025
n8483.1	Tap the Pequest River-Washington 34.5 kV line, approximately 2.2 miles from Pequest River and 5.6 miles from Washington. In addition, install two line switches with SCADA control. And provide 120 V power to the new switch structures.	0.7921	12/17/2021
n8483.2	Upgrade relay settings at Pequest River 34.5 kV substation.	0.0753	12/17/2021
n8483.3	Upgrade relay settings Washington 34.5 kV substation.	0.0753	12/17/2021
n8485.1	Build a three-breaker AE2-051 500 kV switching substation.	19.5	6/30/2027
n8485.2	Rearrange line No. 562 to loop into and out of the new three-breaker AE2-051 500 kV switching station.	3.85	6/30/2027
n8485.3	Remote protection and communications work at Septa and Carson 500 kV substations	0.196	6/30/2027
n8486.1	Install physical security devices and other construction work by FirstEnergy at new AE2-139 interconnection substation by developer.	1.1368	5/31/2027
n8486.2	Estimated design, installation and testing/commissioning for MPLS equipment for SCADA transport at AE2-139 230 kV interconnection substation	0.2374	5/31/2027
n8486.3	Install ADSS from East Towanda 230 kV substation to the new AE2-139 230 kV interconnection substation.	0.4388	5/31/2027
n8486.4	East Towanda-Marshall 230 kV: Cut the existing East Towanda-Marshall 230 kV line approximately 4 miles from the Scotch Hollow tap to create a loop into the new three-breaker ring bus for AE2-139 interconnection.	1.5592	5/31/2027
n8486.5	Marshall 230 kV substation: Modify relay settings, drawings review and nameplate.	0.0742	5/31/2027

n8486.6	East Towanda 230 kV substation: Modify relay settings, drawings review and nameplate.	0.0734	5/31/2027
n8486.7	Scotch Hollow 230 kV substation: Modify relay settings, drawings review and nameplate.	0.0747	5/31/2027
n8486.8	Grover 230 kV substation: Relay settings, drawings review and nameplate.	0.0744	5/31/2027
n8488	Install 34.5 kV potential transformer and replace Oil Creek phase and RC relays with SEL-351S at Titusville 34.5 kV substation.	0.649	7/31/2027
n8489.1	Replace 34.5 kV South Troy line relaying and controls and install one 34.5 kV PT.	0.5466	7/31/2027
n8489.2	Revise relay settings at South Troy 34.5 kV substation.	0.0568	7/31/2027
n8492	Construct one overhead 500 kV transmission line that will start at the existing Fentress 500 kV substation and terminate at the existing Yadkin 500 kV substation, located approximately 13.7 miles away.	73.7224	12/31/2025
n8492.1	Add one new 500 kV breaker position and associated equipment at the Fentress 500 kV substation to terminate the new Yadkin-Fentress 500 kV line.	99.5838	12/31/2025
n8492.2	Add one new 500 kV breaker position and associated equipment at the Yadkin 500 kV substation and relocate existing line No. 565 (Suffolk-Yadkin) as necessary to accommodate the construction of new Yadkin-Fentress 500 kV line.	15.4465	12/31/2025
n8496.1	Lucerne 115 kV substation: Convert Lucerne substation to a four-breaker (future six breaker) bus.	10.4961	5/31/2027
n8496.2	Lucerne-Shelocta 115 kV line: Reterminate the Lucerne-Shelocta 115 kV line to accommodate the new Lucerne 115 kV substation layout.	0.6392	5/31/2027
n8496.3	East Pike-Lucerne 115 kV line: Reterminate the East Pike-Lucerne 115 kV line to accommodate the new Lucerne 115 kV substation layout.	1.0317	5/31/2027
n8496.4	East Pike 115 kV substation: Replace one 115 kV line trap and line tuner with one wideband line trap and line tuner for the Glory line terminal.	0.6317	5/31/2027
n8496.5	Glory 115 kV substation: Replace one 115 kV line trap and line tuner with one wideband line trap and tuner.	0.6367	5/31/2027
n8496.6	Shelocta 115 kV: Install one PCM 5350 for the Lucerne 115 kV line.	0.4608	5/31/2027
n8497	Revise relay settings as needed on 34.5 kV Turner Street line.	0.0805	8/29/2027
n8498.1	AF2-187 interconnection substation: Install physical security and camera system.	0.389042	9/30/2027
n8498.2	Cloverdale-East Wooster 138 kV line: Install tangent structure to support the Cloverdale-East Wooster 138 kV circuit parallel to the Brookside-Cloverdale 138 kV loop location.	0.227552	9/30/2027
n8498.3	Brookside-Cloverdale 138 kV Loop: Loop the Brookside-Cloverdale 138 kV approximately 18.3 miles from the Cloverdale 138 kV substation near structure No. 487 into the new AF2-187 interconnection substation.	0.8254	9/30/2027
n8498.4	Ross substation 138 kV: Install in-sub fiber run and SEL-2506.	0.427003	9/30/2027
n8498.5	Brookside substation 138 kV: Install new wave trap, line-tuning unit and revise relay settings.	0.77921	9/30/2027
n8498.6	Cloverdale substation 138 kV: Install new wave trap, line-tuning unit and revise relay settings.	0.747618	9/30/2027

n8498.7	AF2-187 138 kV interconnection substation: Design, installation and testing/commissioning for MPLS equipment for SCADA transport at AF2-187 interconnection substation.	0.327978	9/30/2027
n8498.8	AF2-187 138 kV interconnection substation: New ADSS fiber installation between AF2-187 interconnection substation and fiber backbone	0.272713	9/30/2027
n8499	Install an adjacent structure on the existing E707 Kittatinny-Newton 34.5 kV line in order to accommodate the new tap installed on the M715 Kittatinny-Newton 34.5 kV line. Install two SCADA-controlled in-line switches and extend distribution circuit to provide power to the switches.	2.7148	8/30/2027
n8504	Expansion of existing Avon 138 kV substation	5.006	4/30/2027
n8505.1	Install two new fiber optic cable paths consisting of 3 miles of 144 count all-dielectric self-supporting (ADSS) cable installed on existing AEP structures, 0.2 miles of 144 count ADLT cable installed in new underground right-of-way, and associated terminating equipment and devices between the proposed AE2-298 switching station, Haviland and South Van Wert 69 kV stations.	0.38673	12/30/2026
n8505.2	Install two new steel, 75' single circuit, single pole dead-end structures on concrete piers with anchor bolt cages and two spans of drake transmission line conductor with seven No. 10 alumoweld shield wire on the Cavett Switch-West Van Wert 69 kV line.	1.006408	12/30/2026
n8505.3	Replace the protective relays and circuit breaker controls, replace the existing coupling capacitor voltage transformer (CCVT) with a new three-phase CCVT and install a new surge arrester at Haviland 69 kV.	0.46964	12/30/2026
n8505.4	Replace the protective relays and circuit breaker controls, replace the existing coupling capacitor voltage transformer (CCVT) with a new three-phase CCVT, and install a new surge arrester at South Van Wert 69 kV.	0.519595	12/30/2026
n8506	Tap line 149 into AE2-259 115 kV switching station.	6.070526	3/1/2030
n8507	Remote protection and communication work at Willis Mountain 115 kV substation	0.217299	3/1/2030
n8511	Install new relays at Bergen 138 kV to accommodate the AF2-415 generator lead line.	0.076	12/31/2026
n8514.1	Install two 34.5 kV load-break air switches with SCADA control on the Larrabee-Point Pleasant B106 34.5 kV line approximately 1.9 miles from Larrabee and 3.9 miles from Allaire.	0.801415	12/31/2026
n8514.2	Revise relay settings at Larrabee 34.5 kV.	0.066673	12/31/2026
n8514.3	Revise relay settings at Point Pleasant 34.5 kV.	0.066673	12/31/2026
n8515	Modify Relay Settings at GM Lordstown, Niles, Magellan, Newton Falls, Evergreen, Mahoningside, Bluebell, Darrow, Shalersville, Lordstown, Salt Springs, Bruce Mansfield, Juniper, Brady, Beaver Valley, Chamberlin, Ravenna and East Akron 345 kV substations and integrate customer protection and controls to the FirstEnergy transmission system.	1.032442	8/12/2025
n8534	Relocation of Auburn-Varner 138 kV structure	0.151627	5/31/2022
n8540	Expansion of existing substation at Southwest Lima 345 kV	1.789243	6/30/2022
n8541	Replace two existing 85' single pole, single circuit tangent structures on the Southwest Lima to West Moulton 138 kV line.	0.311142	6/30/2022
n8594.0	Replace all 447 all aluminum conductors, replace the bus differential panel, review and revise the protective relay settings at Stone Lake 69 kV.	0.730618	7/22/2026

n8595.0	Review and revise relays at Kenzie Creek 69 kV.	0.071413	7/22/2026
n8596.0	Install two new fiber optic cable paths at Kenzie Creek-Stone Lake 69 kV line.	0.238921	7/22/2026
n8597.0	Rebuild 0.6 miles of the Pokagon-Colby 69 kV circuit.	2.19111	7/22/2026
n8598.0	Removal of equipment at Pokagon-Colby 69 kV circuit.	0.37206	7/22/2026
n8599.0	Construct a new 69 kV ring bus station, initially populated with three circuit breakers, expandable to four breakers on the Kenzie Creek-Stone Lake 69 kV circuit.	7.289589	7/22/2026
n8600.0	Install of two new steel, 85' single circuit, single pole dead-end structures on concrete piers with anchor bolt cages and two spans of ACSR 1033.5 (Curlew) transmission line conductor with 7 No. 8 Alumoweld shield wire on the Kenzie Creek-Stone Lake 69 kV circuit.	1.200325	7/22/2026

Revision History

1. Original posting
2. Change the TO field in Attachment A and B tables to reflect the correct entities including POTES, VEPCO and VLT
3. Corrected the Feb. 26, 2025 board approval date in the Executive Summary